SUMS OF TWO SQUARES WITH FEW PRIME FACTORS VIA THETA SERIES OF QUATERNARY FORMS

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ABSTRACT. Let $N \equiv 2 \pmod 4$ be large. We prove that the number of $n < N$ such that n and $N-n$ are both sums of two squares, $\Omega(n(N-n)) \le 8$ and $\max(\Omega(n), \Omega(N-n)) \le 5$ is $\gg N/(\log N)^2$; the two bounds hold simultaneously for the same n. The previous bounds, obtained by sifting the primes $\equiv 3 \pmod 4$ and switching, were 11 and 8. The new input is a level of distribution $1/2$ for the weighted sequence $r(n)r(N-n)$, where r counts representations as a sum of two squares. For coprime m, m' the sum $\sum_{ma+m'b=N} r(a)r(b)$ is the N-th coefficient of the weight-two theta series of $m(x^2+y^2)+m'(z^2+w^2)$. We prove that its cuspidal part $G_{m,m'}$ has Petersson norm $\|G_{m,m'}\|^2 \ll (mm')^{\varepsilon}$, by computing with Waibel's method the expansion of the theta series at every cusp of $\Gamma_0(4mm')$ (Waibel's general theorem gives $(mm')^{1/2+\varepsilon}$); combining this with Deligne's bound through the Hecke operators, and reducing the non-coprime cases by the operators U_p, we obtain $\sum_{ma+m'b=N} r(a)r(b) = (\text{Siegel main term}) + O(d(N)N^{1/2}(mm')^{\varepsilon})$ uniformly in m, m', which is the level $1/2$. Waibel's bound alone gives the level $2/5$ and the bounds 9 and 6. Numerical experiments over a finite range show that the error does not grow with mm' at all, and that its root mean square decreases like $(\log mm')(mm')^{-1/2}$.

1. INTRODUCTION

Let $r(n) = \#\{(x,y) \in \mathbb{Z}^2 : \ x^2+y^2 = n\}$, so that $r(n) = 4\sum_{d|n} \chi(d)$ with $\chi = \chi_{-4}$ the non-trivial character modulo 4. Every integer N is a sum $n + (N-n)$ of two sums of two squares, and in [3] we asked for such representations in which n and $N-n$ have few prime factors. For $N \equiv 2 \pmod 4$ it was shown there, by a vector sieve in which the primes $q \equiv 3 \pmod 4$ are sifted up to nearly $N^{1/2}$ and the survivors with two large such primes are removed by a switching argument, that there are $\gg N/(\log N)^2$ integers $n < N$ with n, $N-n$ sums of two squares and $\Omega(n(N-n)) \le 11$, and $\gg N/(\log N)^2$ with $\max(\Omega(n), \Omega(N-n)) \le 8$.

Here we take a different route: instead of sifting the primes $\equiv 3 \pmod 4$ we count with the weight $r(n)r(N-n)$, which vanishes unless both n and $N-n$ are sums of two squares. The price is that the sequence $r(n)r(N-n)$ must be shown to be well distributed in arithmetic progressions, and the gain is that no inert prime has to be sifted and no switching is needed. We prove a level of distribution $1/2$ (Theorem 3.2), and the sieve then gives the following.

Theorem 1.1. *Let $N \equiv 2 \pmod 4$ be sufficiently large. Then*

$$\#\big\{1 \le n < N : \ n = x^2+y^2, \ N-n = z^2+w^2, \ \Omega(n(N-n)) \le 8, \ \max(\Omega(n), \Omega(N-n)) \le 5\big\} \ge \ c_1 \, \mathfrak{S}_r(N) \, \frac{N}{(\log N)^2},$$

where $c_1 > 0$ is an absolute constant and

$$\mathfrak{S}_r(N) = \prod_{\substack{p|N \\ p>2}} \frac{p-1}{p-2} \tag{1}$$

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is the singular series of the binary Goldbach problem; in particular $\mathfrak{S}_r(N) \geq 1$. *The integers counted are coprime to* N, *and each of* n, $N-n$ *is a product of distinct primes* $\equiv 1 \pmod 4$ *exceeding* $N^{6/125}$.

The constant can be made explicit, $c_1 = 0.23$ being admissible (Section 4); we have not tried to optimise it. That the arithmetic factor is the Goldbach singular series is explained in Section 3: under the weight $r(n)r(N-n)$ the events $p \mid n$ and $p \mid N-n$ have the same density $h(p)$ and are mutually exclusive, so the density of $p \nmid n(N-n)$ is $1-2h(p)$, and $(p-1)/(p-2)$ is the ratio of the local factor at a prime dividing N to this generic factor. Both bounds 8 and 5 come from one set of sieve parameters, so they hold simultaneously for the same integers n. In Part II [4, Theorem 7.1] the vector sieve of Section 4 is replaced by the two-dimensional sieve of Diamond, Halberstam and Richert, which improves the bound 8 to 7 (with 5 unchanged) from the same level of distribution $1/2$; the sieve inequality behind it is certified in interval arithmetic.

The analytic input is a bound for the Petersson norm of the cuspidal part $G_{m,m'}$ of the theta series $\theta^2(mz)\theta^2(m'z)$ of the quaternary form $Q_{m,m'} = m(x^2+y^2) + m'(z^2+w^2)$, with the inner product (4) below.

Theorem 1.2 (The norm)**.** *For coprime odd positive integers* m, m', $\|G_{m,m'}\|^2 \ll_\varepsilon (mm')^\varepsilon$.

Waibel's uniform theorem [18, Theorem 1] gives $\|G_{m,m'}\|^2 \ll (mm')^{1/2+\varepsilon}$ for these forms (Lemma 2.1), which is sharp for general quaternary forms. Theorem 1.2 is proved in Section 2.5 by running Waibel's method without his general lattice-point estimate: his decomposition of the norm over the cusps of $\Gamma_0(4mm')$ and his description of the expansion of a theta series at a cusp show that, for $Q_{m,m'}$, the expansion at every cusp is again a theta series of a form $A(x_1^2+x_2^2) + B(x_3^2+x_4^2)$ with $AB \asymp mm'$, whose coefficients are $O(n^\varepsilon)$ in the range $n \ll mm'$ that matters, because the equation $A\alpha + B\beta = n$ has $O(1)$ solutions. The present paper used only Waibel's bound in its first two versions; everything below is written so that the reader can see what Lemma 2.1 alone gives (the level $2/5$ and the bounds 9 and 6, Remark 4.5) and what Theorem 1.2 adds. Part II [4] contains the mean square of $T-M$ over N, the level of distribution $1-\varepsilon$ for almost all N, and the numerical measurement of the norm ($\|G_{m,m'}\|^2 \approx 0.37(\log mm')^2$).

Summary of contributions. We separate what is proved here from what is taken from the literature, and what is proved from what is conjectured.

(a) *The arithmetic result.* Theorem 1.1 improves the bounds 11 and 8 of [3] to 8 and 5, with the same order of magnitude $N/(\log N)^2$ for the count, and with both bounds holding for the same integers.

(b) *A uniform coefficient bound.* The analytic inputs are the norm bound of Theorem 1.2, proved in Section 2.5 with the tools of Waibel [18] (his decomposition of the norm over the cusps and his expansion of a theta series at a cusp), and Deligne's theorem [5]. A norm bound does not by itself control a single Fourier coefficient uniformly in the level: the Petersson formula applied at $n = N$ loses a factor $N^{1/4}$ when the level is small compared with $N^{1/2}$. Our contribution here is the passage from the norm to the N-th coefficient uniformly in m, m': the part of N coprime to the level is moved into a Hecke operator, whose operator norm is controlled by Deligne's bound (Lemma 2.2), and the primes of mm' dividing N are removed by an exact recursion with the operators U_p whose coefficients we count (Lemma 2.5). The result is the uniform estimate (3) below, in which the dependence on mm' is only $(mm')^\varepsilon$.

(c) *The sieve set-up.* We prove the level of distribution $1/2$ for the weighted sequence $r(n)r(N-n)$ (Theorem 3.2). With this weight the inert primes have density $\ll q^{-2}$ and need not be sifted, so the sieve is a two-dimensional vector sieve of two linear components, without the switching step of [3], and the final numerical inequality is correspondingly simpler (Section 4, Appendix A). The two components are not independent, the events $p \mid n$ and $p \mid N-n$

being mutually exclusive; the small primes are therefore sifted from the product $n(N-n)$ by the fundamental lemma and only the large ones by the vector sieve (Proposition 4.3).

(d) *Logical status.* Theorem 1.1, the norm bound (Theorem 1.2), the uniform bound (3) and the level $1/2$ (Theorem 3.2) are unconditional and proved here. With Waibel's bound (Lemma 2.1) in place of Theorem 1.2, the same proofs give the error term $O(d(N)N^{1/2}(mm')^{1/4+\varepsilon})$, the level $2/5$ and the bounds 9 and 6 (Remark 4.5); this was the content of the first two versions of this paper. Numerical: the experiments of Section 5 measure how the error term depends on mm' over a finite range and enter no proof. The only computer-assisted step in a proof is the verification of the inequality $L-\lambda R>0$ of Section 4, which is carried out in interval arithmetic.

Where the gain comes from. Table 1 lists, step by step, what is taken from the literature and what is added here. Three inputs are new: the norm bound (Theorem 1.2), the passage from the norm to a coefficient (Lemma 2.2), and the sieve structure. Their shares in the final result can be read off the sieve itself, since the numerical inequality depends on the level θ only: with the sieve of Section 4, the level $1/3$—which is what the Petersson formula applied directly at $n=N$ yields from Waibel's bound—gives $\Omega(n(N-n))\le 10$ and $\max\Omega\le 6$, the level $2/5$ (Waibel's bound and Lemma 2.2) gives 9 and 6, and the level $1/2$ (Theorem 1.2 and Lemma 2.2) gives 8 and 5 (floating-point scan with `modular_sieve_scan.py`; the last two pairs are certified, Table 2). The remaining distance to the bounds 11 and 8 of [3] is due to the change of weight: with $r(n)r(N-n)$ no inert prime has to be sifted and no switching argument is needed.

Step	From the literature	Added here
Main term	Siegel's theorem: the Eisenstein part of $\theta^2(mz)\theta^2(m'z)$ has coefficients $\prod_p \beta_p$ [16, 13, 18]	the densities (6)–(7), the 2-adic equivalence $Q_{m,m'}\cong Q_{1,1}$ for $m\equiv m'\equiv 1$ (4), and the main term $Xh(d)h(e)$ of the sieve (Lemma 3.1)
Cusp part, norm	$\|G_{m,m'}\|^2\ll (mm')^{1/2+\varepsilon}$ [18, Thm. 1]; decomposition of the norm over the cusps and expansion of a theta series at a cusp [18, (22), Lemma 12]	Theorem 1.2 (Section 2.5): $\|G_{m,m'}\|^2\ll (mm')^{\varepsilon}$, because for $Q_{m,m'}$ the expansion at every cusp is a theta series of $A(x_1^2+x_2^2)+B(x_3^2+x_4^2)$ with $AB\asymp mm'$
Norm → coefficient	Petersson formula at $n=N$: $\|a_G(N)\|\ll \|G\|N^{1/2+\varepsilon}(1+N^{1/4}(mm')^{-1/2})$, i.e. level of distribution $1/3$ with Waibel's bound	Lemma 2.2: $\|a_G(N)\|\le C\,d(N)N^{1/2}\|G\|$ via T_{n_1} and Deligne's bound, i.e. level $2/5$ with Waibel's bound and $1/2$ with Theorem 1.2 (Remark 3.3)
$(N,mm')>1$	—	Lemma 2.5: exact reduction by U_p, coefficient total $\le 4^{\omega(N)}d(N)$; needed for the Möbius inversion over the divisors of N in (16)
Sieve	[3]: unweighted sequence, inert primes sifted up to $N^{1/2-\varepsilon}$, switching; bounds 11, 8	weight $r(n)r(N-n)$, inert primes of density q^{-2}, fundamental lemma on $n(N-n)$ plus vector sieve of two linear components with Richert weights; bounds 8, 5 for the same n (9, 6 at the level $2/5$)

TABLE 1. Inputs from the literature and contributions of this paper.

Outline of the proof. The proof has five steps, which Sections 2–4 carry out in this order.

(1) *From the sieve to theta coefficients.* Write a square-free d as a product of primes $p \equiv 1 \pmod 4$ (split) and $q \equiv 3 \pmod 4$ (inert). The identities $r(n) = 2r(n/p) - r(n/p^2)$ for split $p \mid n$ and $r(n) = r(n/q^2)$ for inert $q \mid n$ express the sum of $r(n)r(N-n)$ over $n \equiv 0\ (d)$, $n \equiv N\ (e)$ as a combination of the sums

(2) $$T(m, m'; N) = \sum_{ma+m'b=N} r(a)r(b) = \#\{(x, y, z, w) \in \mathbb{Z}^4 :\ m(x^2+y^2) + m'(z^2+w^2) = N\}$$

with $m \mid d^2$, $m' \mid e^2$. This is the number of representations of N by the positive quaternary form $Q_{m,m'} = m(x^2+y^2) + m'(z^2+w^2)$, i.e. the N-th Fourier coefficient of the theta series $\theta^2(mz)\theta^2(m'z)$, a holomorphic modular form of weight 2 on $\Gamma_0(4mm')$. By Siegel's theorem it is the sum of the N-th coefficient of an Eisenstein series, which is the product $M(m, m'; N)$ of the local densities, and of the N-th coefficient $a_G(N)$ of a cusp form $G = G_{m,m'}$. The main terms M recombine into the product form $Xh(d)h(e)$ required by the sieve (Section 3).

(2) *The norm of the cusp part.* Deligne's bound controls $a_G(N)$ for each fixed form, and the whole difficulty is the uniformity in the level $4mm'$. Waibel's theorem [18] gives $\|G_{m,m'}\|^2 \ll (mm')^{1/2+\varepsilon}$ for our forms (Lemma 2.1); Theorem 1.2, proved in Section 2.5 with Waibel's cusp decomposition, gives $\|G_{m,m'}\|^2 \ll (mm')^{\varepsilon}$.

(3) *From the norm to one coefficient.* The Petersson formula applied at the index N loses a factor when the level is small compared with $N^{1/2}$. We write $N = n_1 n_2$ with n_1 coprime to the level and move n_1 into the Hecke operator T_{n_1}, which acts on each oldform space as a scalar bounded by Deligne's theorem; the Cauchy–Schwarz step then needs the Petersson formula only at the index $n_2 \in \{1, 2\}$, where its off-diagonal term is negligible (Lemma 2.2). For $(N, mm') = 1$ this gives $T = M + O(d(N)N^{1/2}(mm')^{\varepsilon})$ (Theorem 2.3).

(4) *Primes of mm' dividing N.* The sieve needs $(n, N) = 1$, and removing this condition by Möbius inversion introduces forms $Q_{m,m'}$ in which primes of mm' divide N, to which Theorem 2.3 does not apply. The operators U_p reduce these terms exactly to the coprime case, with an integer combination whose coefficients total at most $4^{\omega(N)}d(N)$ (Lemma 2.5), and we obtain

(3) $$T(m, m'; N) = M(m, m'; N) + O_{\varepsilon}\big(d(N)^4 N^{1/2}(mm')^{\varepsilon}\big)$$

uniformly in coprime odd cube-free m, m' (Corollary 2.7; for $(N, mm') = 1$ the factor $d(N)^4$ can be replaced by $d(N)$).

(5) *Level of distribution and sieve.* Since the main term is of size $N/(mm')$, the error in (3) is smaller than the main term as long as $mm' \le N^{1/2-\varepsilon}$, and a bookkeeping argument over the moduli turns (3) into the level of distribution $1/2$ for $r(n)r(N-n)$ (Theorem 3.2). Under the weight $r(n)r(N-n)$ a split prime p divides n with density $h(p) = (2p-1)/(p(p+1)) \approx 2/p$ and an inert prime only with density $1/(q(q-1))$; so sifting all primes below N^{α} on both sides is a vector sieve of two linear components, with no inert prime to sift and no switching. Richert's weights then bound $\Omega(n(N-n))$ and $\max(\Omega(n), \Omega(N-n))$ through one positivity condition $L - \lambda R > 0$, which is verified with interval arithmetic (Section 4 and Appendix A); since one set of weights gives both bounds, they hold for the same n.

Relation to the shifted convolution literature. The sum (2) is a shifted convolution problem for the coefficients of θ^2 with linear coefficients m, m', in the "sum" orientation $ma + m'b = N$. For the divisor function in place of r and the "difference" orientation $r_1 n - r_2 m = h$, Topacogullari [17] proved an asymptotic formula uniform in the coefficients, with an error term of the shape $(r_2 x_1)^{1/2+\vartheta+\varepsilon}$ (in his notation), where ϑ is any admissible exponent towards the Ramanujan–Petersson conjecture for Maass forms, $\vartheta = 7/64$ by Kim and Sarnak [12]. The two problems are structurally different, and we do not regard one result as an improvement of the other.

In the difference orientation the natural spectral expansion is over the full L^2 spectrum of Γ_0 of the relevant level, including Maass cusp forms, whose possible exceptional eigenvalues and non-tempered local components enter through ϑ. In the sum orientation, $T(m,m';N)$ is a Fourier coefficient of a *holomorphic* modular form of weight 2, so only holomorphic cusp forms occur, Deligne's theorem gives the Ramanujan–Petersson bound at all finite places, and the uniformity in the level must instead come from a bound for the Petersson norm (Theorem 1.2). The error term in (3) is then Deligne's bound for a single form, with no loss in the level beyond $(mm')^{\varepsilon}$: the Cauchy–Schwarz step in the passage from the norm to a single coefficient (Lemma 2.2) costs exactly the factor $\|G_{m,m'}\|$, which is at most $(mm')^{\varepsilon}$. Numerically (Section 5) the quantity $\sup_N |T-M|/\sqrt{N}$ does not grow with mm' at all, while the root mean square decreases like $(\log mm')(mm')^{-1/2}$, in agreement with Theorem 1.2 and the Rankin–Selberg method; Part II proves the corresponding mean-square bound and derives from it the level of distribution $1-\varepsilon$ for almost all N.

Notation. p denotes a prime $\equiv 1 \pmod 4$ and q a prime $\equiv 3 \pmod 4$ when the distinction matters; $\chi = \chi_{-4}$; $d(n)$ and $\omega(n)$, $\Omega(n)$ are the usual divisor and prime-factor counting functions, $d_3(n)$ the ternary divisor function; $\sigma_{-1}(n) = \sum_{t\mid n} 1/t$. The Petersson inner product on $S_2(\Gamma_0(L))$ is

$$\langle f, g\rangle = \int_{\Gamma_0(L)\backslash\mathbb{H}} f(z)\overline{g(z)}\,\mathrm{d}x\,\mathrm{d}y, \qquad \|f\|^2 = \langle f, f\rangle, \tag{4}$$

which is [8, (2.1)] and the normalisation of [18] for weight 2. All implied constants may depend on ε.

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2. Theta series of the forms $m(x^2+y^2)+m'(z^2+w^2)$

Throughout this section m, m' are coprime odd positive integers, $Q = Q_{m,m'}$ is the quaternary form $m(x^2+y^2)+m'(z^2+w^2)$ with Gram matrix $A = \operatorname{diag}(2m, 2m, 2m', 2m')$, and

$$\Theta(z) = \theta^2(mz)\theta^2(m'z) = \sum_{n\ge 0} T(m,m';n)e(nz), \qquad \theta(z) = \sum_{k\in\mathbb{Z}} e(k^2 z).$$

Since $\theta^2 \in M_1(\Gamma_0(4), \chi)$, we have $\Theta \in M_2(\Gamma_0(4mm'))$ with trivial character. In the terminology of [18], Q is primitive (because $(m,m')=1$), $\det A = 16(mm')^2$, and the level of Q, the least L such that LA^{-1} is integral with even diagonal, is $L = 4mm'$.

2.1. Siegel's theorem and the local densities.

By Siegel's theorem [16] (for weight 2 see Kitaoka [13] and [18, Section 2]), $\Theta = E + G$ where E is the theta series of the genus of Q (a linear combination of Eisenstein series) and $G \in S_2(\Gamma_0(4mm'))$ is a cusp form, and the coefficients of E are given by the product of local densities [18, (1)–(2)]

$$M(m,m';n) = \frac{(2\pi)^2 n}{\Gamma(2)\sqrt{\det A}} \prod_p \beta_p(n) = \frac{\pi^2 n}{mm'} \prod_p \beta_p(n), \tag{5}$$

where $\beta_p(n) = \lim_{a\to\infty} p^{-3a}\#\{x \in (\mathbb{Z}/p^a)^4 :\ Q(x) \equiv n\ (p^a)\}$. We record the densities that we need. For $p \nmid 2mm'$ the form is equivalent over $\mathbb{Z}_p$ to $x^2+y^2+z^2+w^2$ and

$$\beta_p(n) = (1-p^{-2})\,\sigma_{-1}(p^{v_p(n)}), \tag{6}$$

as follows from Jacobi's formula $r_4(n) = 8\sum_{t|n,\,4\nmid t} t$ (or from a direct count). For $p \mid mm'$ with $p \nmid n$ we have

$$\beta_p(n) = 1 - \frac{\chi(p)}{p}\,: \tag{7}$$

indeed, if $p \mid m$ then modulo p the equation reads $m'(z^2+w^2) \equiv n$, which has $p - \chi(p)$ solutions (z,w), none of them with $z \equiv w \equiv 0$; every such solution is non-singular and lifts to $p^{3(a-1)}$ solutions modulo p^a by Hensel's lemma, so $\beta_p(n) = p^2(p-\chi(p))/p^3$. At $p = 2$ the density $\beta_2(n) = \beta_2(Q_{m,m'}; n)$ depends on m, m' only through their residues modulo 4, and it does depend on them. If $m \equiv m' \equiv 1 \pmod 4$, then $Q_{m,m'}$ is equivalent over $\mathbb{Z}_2$ to $Q_{1,1} = x^2+y^2+z^2+w^2$: the norms of the units of $\mathbb{Z}_2[i]$ form the subgroup $1 + 4\mathbb{Z}_2$ of $\mathbb{Z}_2^\times$ (a norm x^2+y^2 of a unit has one of x, y even and is $\equiv 1 \pmod 4$, and the subgroup has index two), so $m = u\bar{u}$ for a unit u, and $x + iy \mapsto u(x+iy)$ transforms x^2+y^2 into $m(x^2+y^2)$. Hence

$$\beta_2(Q_{m,m'}; n) = \beta_2(Q_{1,1}; n) \quad (m \equiv m' \equiv 1\ (4)), \qquad \beta_2(Q_{1,1}; n) = \begin{cases} 1, & n \text{ odd}, \\ 3/2, & n \equiv 2\ (4), \end{cases} \tag{8}$$

the values of $\beta_2(Q_{1,1}; n)$ following from Jacobi's formula, since $M(1,1;n) = r_4(n)$ exactly (there are no cusp forms of level 4). The same holds for $m \equiv m' \equiv 3 \pmod 4$, since then $Q_{m,m'} \cong -Q_{1,1}$ over $\mathbb{Z}_2$ and $\beta_2(Q_{1,1}; n)$ depends only on $v_2(n)$. For $m \not\equiv m' \pmod 4$, however, $\beta_2(Q_{m,m'}; n) = 1/2$ when $n \equiv 2 \pmod 4$ (the representations with x^2+y^2 and z^2+w^2 both odd are absent, as $m + m' \equiv 0 \pmod 4$; one checks the value by counting solutions modulo 2^6), while it is still 1 for odd n. We shall use (8) only for $m \equiv m' \equiv 1 \pmod 4$, which is the only case arising in Section 3, and otherwise only the fact that $\beta_2(n) \asymp 1$ for $v_2(n) \le 1$. In particular, for $(n, mm') = 1$ and $m \equiv m' \pmod 4$,

$$\frac{M(m,m';n)}{M(1,1;n)} = \frac{1}{mm'} \prod_{p \mid mm'} \frac{1-\chi(p)/p}{1-p^{-2}} = \frac{1}{mm'} \prod_{p \mid mm'} g(p), \tag{9}$$

where $g(p) = p/(p+1)$ for $p \equiv 1 \pmod 4$ and $g(q) = q/(q-1)$ for $q \equiv 3 \pmod 4$. Since $\beta_p(n) \le (1-p^{-2})\sigma_{-1}(p^{v_p(n)})$ for $p \nmid 2mm'$ and $\beta_p(n) \le 4/3$ for $p \mid mm'$, $p \nmid n$, (5) gives the trivial bound

$$M(m,m';n) \ll \frac{n}{mm'}\,\sigma_{-1}(n) \prod_{p \mid mm'} \Big(1 + \frac{1}{p}\Big) \ll_\eta \frac{n^{1+\eta}}{mm'} \qquad \big((n, mm') = 1,\ v_2(n) \le 1\big). \tag{10}$$

2.2. **Waibel's bound.**

Lemma 2.1. $\|G\| \ll (mm')^{1/4+\varepsilon}$.

Proof. Theorem 1 of [18] states, for a primitive integral positive quaternary form of level L and $f = \theta(Q) - \theta(\mathrm{gen}Q)$,

$$\langle f, f\rangle \ll \frac{L^{2+\varepsilon}}{F(Q,2)} + \frac{L^{1+\varepsilon}}{(\det A)^{1/4}},$$

with the inner product (4) and a genus invariant $F(Q,s)$ defined in [18, (6)] from the Jordan decompositions of Q over $\mathbb{Z}_p$: $F(Q,s) = 2^{\min(4-2r_2,s)} \prod_{p|L} p^{\mu_p(s)}$, $\mu_p(s) = \sum_{j=1}^{v_p(L)} \min(v_p(j), s)$, where $v_p(j)$ is the number of variables whose p-adic coefficient has valuation $\ge j$ and r_2 the number of binary 2-adic blocks. For $Q_{m,m'}$ and an odd prime p with $p^a \,\|\, mm'$ the p-adic valuations of the coefficients are $(a, a, 0, 0)$, so $v_p(j) = 2$ for $j \le a$ and $\mu_p(2) = 2a$; at $p = 2$ all coefficients are units, $r_2 = 0$ and $\mu_2(2) = 0$. Hence $F(Q,2) = 4(mm')^2$ (and $F(Q,4) = 16(mm')^2 = \det A$, as it must).

Here $f = G$, since $\theta(\mathrm{gen}Q) = E$ by Siegel's theorem. With $L = 4mm'$ and $\det A = 16(mm')^2$ the two terms are

$$\frac{(4mm')^{2+\varepsilon}}{4(mm')^2} \ll (mm')^{\varepsilon}, \qquad \frac{(4mm')^{1+\varepsilon}}{2(mm')^{1/2}} \ll (mm')^{1/2+\varepsilon},$$

so $\|G\|^2 = \langle G, G\rangle \ll (mm')^{1/2+\varepsilon}$. Taking square roots, $\|G\| \ll (mm')^{1/4+\varepsilon/2} \le (mm')^{1/4+\varepsilon}$. □

Theorem 1.2 replaces the exponent $1/4$ of Lemma 2.1 by ε. Its proof, which uses Waibel's method but none of the material of Sections 2.3–2.4, is given in Section 2.5 at the end of this section; we first show how a bound for the norm is turned into a bound for a single coefficient.

2.3. **From the norm to a single coefficient.** The Petersson formula ([8, Prop. 2.1], [7, Prop. 14.5], stated there for all even $k \ge 2$ and written here for $k = 2$ and the unnormalised coefficients $a_f(n)$) gives, for an orthonormal basis $\mathcal{B}$ of $S_2(\Gamma_0(L))$ with respect to (4) and $n \ge 1$,

$$\sum_{f\in\mathcal{B}} |a_f(n)|^2 = 4\pi n\, \Delta_L(n,n), \qquad \Delta_L(n,n) = 1 - 2\pi \sum_{\substack{c>0\\ c\equiv 0 \ (L)}} \frac{S(n,n;c)}{c}\, J_1\Big(\frac{4\pi n}{c}\Big), \tag{11}$$

where $S(m,n;c)$ is the Kloosterman sum and J_1 the Bessel function; the series converges absolutely by Weil's bound. The off-diagonal sum is negligible when n is small compared with L (Step 4 below), but not at $n = N$: there Weil's bound gives only $\Delta_L(N,N) = 1 + O(N^{1/2+\varepsilon}/L)$ for $(N,L) = 1$ (compare [8, Cor. 2.2], [7, Cor. 14.24], used in this form in [18, (3)]), which costs a factor $1 + N^{1/4+\varepsilon}L^{-1/2}$ for small levels and would give the level of distribution $1/3$ in Section 3 instead of $1/2$ (Remark 3.3). We route the coprime part of N through a Hecke operator instead.

Lemma 2.2. *Let $L \ge 1$, $G \in S_2(\Gamma_0(L))$, and $n = n_1n_2$ with $(n_1, L) = 1$, $n_2 \mid L^\infty$ and $n_2 \le L^{1/2}$. Then*

$$|a_G(n)| \le C\, d(n_1)\, n^{1/2}\, \|G\|$$

with an absolute constant C.

Proof. Step 1: the operator norm of T_{n_1}. For a newform f of level $L' \mid L$ and $t \mid L/L'$ let $B_tf(z) = f(tz)$. By Atkin–Lehner theory [1] (see also [11, Sections 6.6–6.7]) the forms B_tf span $S_2(\Gamma_0(L))$, and

$$S_2(\Gamma_0(L)) = \bigoplus_f S(f), \qquad S(f) = \mathrm{span}\{B_tf : \ t \mid L/L'\},$$

the sum running over the newforms of level dividing L. For $(n_1, L) = 1$ the Hecke operator T_{n_1} on $M_2(\Gamma_0(L))$ acts on q-expansions by

$$a_{T_{n_1}g}(k) = \sum_{s\mid(n_1,k)} s\, a_g\Big(\frac{n_1k}{s^2}\Big), \tag{12}$$

a formula in which L does not appear; so on a newform f of level $L' \mid L$ the level-L and level-L' operators coincide, and $T_{n_1}f = a_f(n_1)f$ with f normalised by $a_f(1) = 1$. Moreover T_{n_1} commutes with B_t when $(t, n_1) = 1$: by (12), the k-th coefficient of $T_{n_1}B_tf$ is $\sum_s s\, a_f(n_1k/(s^2t))$ over the $s \mid (n_1, k)$ with $t \mid n_1k/s^2$, i.e. (as $(t, n_1) = 1$) with $t \mid k/s$, while that of $B_tT_{n_1}f$ is $\sum_s s\, a_f(n_1k/(ts^2))$ over the $s \mid (n_1, k/t)$ (and 0 if $t \nmid k$); since $(s, t) = 1$ for $s \mid n_1$, both index sets are $\{s \mid n_1 : \ st \mid k\}$. Hence $T_{n_1}B_tf = B_tT_{n_1}f = a_f(n_1)B_tf$, and T_{n_1} acts on $S(f)$ as the scalar $a_f(n_1)$. The operators T_n, $(n, L) = 1$, are self-adjoint for (4) (the character is trivial), and two distinct newforms have distinct systems of eigenvalues $\{a_f(n)\}_{(n,L)=1}$ by strong multiplicity one; hence the spaces $S(f)$ are mutually orthogonal. (The forms B_tf inside one $S(f)$ need not

be orthogonal, but this plays no role, since T_{n_1} is a scalar on $S(f)$.) Writing $G = \sum_f G_f$ with $G_f \in S(f)$, we get

$$\|T_{n_1} G\|^2 = \sum_f |a_f(n_1)|^2 \|G_f\|^2 \le \max_f |a_f(n_1)|^2 \, \|G\|^2.$$

By Deligne's theorem [5], $|a_f(p)| \le 2p^{1/2}$ for every newform f of weight 2 and every prime $p \nmid L'$; the Hecke relations then give $|a_f(p^j)| \le (j+1)p^{j/2}$ and, by multiplicativity, $|a_f(n_1)| \le d(n_1) n_1^{1/2}$ for $(n_1, L') = 1$. Hence $\|T_{n_1} G\| \le d(n_1) n_1^{1/2} \|G\|$.

Step 2: the coefficient. For $H = T_{n_1} G$, formula (12) gives $a_H(n_2) = \sum_{s \mid (n_1, n_2)} s \, a_G(n_1 n_2 / s^2)$. Since $(n_1, n_2) = 1$ only $s = 1$ occurs, so $a_H(n_2) = a_G(n)$.

Step 3: Cauchy–Schwarz. Expanding $H = \sum_{f \in \mathcal{B}} \langle H, f \rangle f$ in an orthonormal basis and using (11),

$$|a_H(n_2)|^2 \le \sum_{f \in \mathcal{B}} |\langle H, f \rangle|^2 \sum_{f \in \mathcal{B}} |a_f(n_2)|^2 = 4\pi n_2 \, \Delta_L(n_2, n_2) \, \|H\|^2.$$

Step 4: the off-diagonal term in Δ_L. By Weil's bound $|S(n, n; c)| \le d(c) \, (n, c)^{1/2} c^{1/2} \le d(c) \, n^{1/2} c^{1/2}$ [7, Cor. 11.12] and $|J_1(x)| \le x/2$ for $x > 0$, the off-diagonal term of (11) at $n = n_2$ is at most

$$2\pi \sum_{\ell \ge 1} \frac{d(L\ell) \, n_2^{1/2} (L\ell)^{1/2}}{L\ell} \cdot \frac{2\pi n_2}{L\ell} \le 4\pi^2 \, d(L) \, \frac{n_2^{3/2}}{L^{3/2}} \sum_{\ell \ge 1} \frac{d(\ell)}{\ell^{3/2}} = 4\pi^2 \zeta(\tfrac{3}{2})^2 \, d(L) \, \frac{n_2^{3/2}}{L^{3/2}},$$

using $d(L\ell) \le d(L) d(\ell)$. The hypothesis $n_2 \le L^{1/2}$ gives $n_2^{3/2} \le L^{3/4}$, and $d(L) \le 2L^{1/2}$, so the off-diagonal term is at most $8\pi^2 \zeta(\frac{3}{2})^2 L^{-1/4}$. Hence $\Delta_L(n_2, n_2) \le C'$ with an absolute constant C' (for fixed n_2, indeed $\Delta_L(n_2, n_2) \to 1$ as $L \to \infty$).

Combining the steps, $|a_G(n)| = |a_H(n_2)| \le (4\pi C' n_2)^{1/2} \, d(n_1) n_1^{1/2} \|G\|$, which is the claim with $C = (4\pi C')^{1/2}$. □

Theorem 2.3. *Let m, m' be coprime odd positive integers and $N \ge 1$ with $(N, mm') = 1$ and $v_2(N) \le 1$. Then*

$$T(m, m'; N) = M(m, m'; N) + O_\varepsilon\big(d(N) \, N^{1/2} (mm')^\varepsilon\big).$$

With Lemma 2.1 in place of Theorem 1.2 the error term is $O_\varepsilon(d(N) N^{1/2} (mm')^{1/4+\varepsilon})$.

Proof. $T - M = a_G(N)$ with $G \in S_2(\Gamma_0(4mm'))$. Apply Lemma 2.2 with $L = 4mm'$, n_1 the odd part of N (coprime to L because $(N, mm') = 1$) and $n_2 = 2^{v_2(N)} \in \{1, 2\}$; then $n_2 \mid L^\infty$ and $n_2 \le 2 \le L^{1/2}$ since $L \ge 4$. Finally $d(n_1) \le d(N)$, and Theorem 1.2, resp. Lemma 2.1, bounds $\|G\|$. □

Remark 2.4. For fixed m, m' the error term $O(d(N) N^{1/2})$ is Deligne's bound and cannot be improved. The point of Theorem 2.3 is that the dependence on mm' is only $(mm')^\varepsilon$; the data of Section 5 show that the supremum of $|T - M| / \sqrt{N}$ over N in a fixed range does not grow with mm' at all.

2.4. **Primes of mm' dividing N.** The sieve also needs $T(m, m'; N)$ when a prime of mm' divides N. Such terms reduce exactly to the coprime case. Let U_p act on q-expansions by $U_p \sum a(n) e(nz) = \sum a(pn) e(nz)$, so that $a_N(\Phi) = a_{N/p}(U_p \Phi)$ when $p \mid N$, and note that $U_p\big(\Phi(pz)\Psi(z)\big) = \Phi(z) \, (U_p \Psi)(z)$. The identities $r(pb) = 2r(b) - r(b/p)$ for split p and $r(qb) = r(b/q)$ for inert q (with $r(x) = 0$ for $x \notin \mathbb{Z}$) say that for $p \nmid m'$

$$U_p \, \theta^2(m'z) = 2\theta^2(m'z) - \theta^2(pm'z) \quad (p \equiv 1 \ (4)), \qquad U_q \, \theta^2(m'z) = \theta^2(qm'z) \quad (q \equiv 3 \ (4)). \tag{13}$$

Lemma 2.5. *Let p be an odd prime with $p \mid N$ and $p \mid m$, $p \nmid m'$.*

(i) If $p \equiv 1$ (4) and $m = pm_1$ with $p \nmid m_1$, then $T(pm_1, m'; N) = 2T(m_1, m'; N/p) - T(m_1, pm'; N/p)$.

(ii) If $p \equiv 1$ (4) and $m = p^2 m_1$, then $T(p^2 m_1, m'; N) = 2T(pm_1, m'; N/p) - T(m_1, m'; N/p^2)$, where the last term is 0 if $p^2 \nmid N$.

(iii) If $p \equiv 3$ (4) and $m = pm_1$ with $p \nmid m_1$, then $T(pm_1, m'; N) = T(m_1, pm'; N/p)$.

(iv) If $p \equiv 3$ (4) and $m = p^2 m_1$, then $T(p^2 m_1, m'; N) = T(m_1, m'; N/p^2)$, which is 0 if $p^2 \nmid N$.

The same holds with the roles of m and m' exchanged. Consequently, if $v_p(mm') \le 2$ for every odd prime p, then $T(m, m'; N)$ is a linear combination with integer coefficients of terms $T(\tilde m, \tilde m'; \tilde N)$ with

(a) *$\tilde m, \tilde m'$ coprime and odd, $v_p(\tilde m \tilde m') \le 2$ for all p, $\tilde m \tilde m' \le mm'$;*

(b) *$\tilde N \mid N$, $v_2(\tilde N) = v_2(N)$, $(\tilde m \tilde m', \tilde N) = 1$ and $\tilde N/(\tilde m \tilde m') \le N/(mm')$;*

the sum of the absolute values of the coefficients being at most $\prod_{p \mid (mm', N)} 4v_p(N) \le 4^{\omega(N)} d(N)$. Since the identities hold for the modular forms themselves, they hold for the Eisenstein parts M and the cusp parts separately.

Proof. The four identities. In case (i), $\theta^2(mz) = \Phi(pz)$ with $\Phi = \theta^2(m_1 \cdot)$, so

$$T(pm_1, m'; N) = a_{N/p}\big(\theta^2(m_1 z)\, U_p \theta^2(m' z)\big),$$

and (13) gives the claim; (iii) is the same with the second identity. In case (ii), $\theta^2(mz) = \Phi(pz)$ with $\Phi = \theta^2(pm_1 \cdot)$, so $T = 2T(pm_1, m'; N/p) - T(pm_1, pm'; N/p)$, and $T(pm_1, pm'; N') = T(m_1, m'; N'/p)$ (or 0 if $p \nmid N'$) because the form is p times $Q_{m_1, m'}$; (iv) is similar, with U_q applied twice.

The recursion at one prime. Fix an odd prime p with $p \mid (mm', N)$. We apply (i)–(iv) repeatedly and keep the following invariant: the current expression is $\sum_j c_j T(m_j, m_j'; N_j)$ in which at most one term, the *active* term, has $p \mid (m_j m_j', N_j)$, and all other terms (*finished*) have $p \nmid (m_j m_j', N_j)$. At the start the only term is active, with coefficient 1. If the active term has $v_p(m_j m_j') = 1$ we apply (i) or (iii), if it has $v_p(m_j m_j') = 2$ we apply (ii) or (iv). Inspecting the right-hand sides:

- (i) replaces an active term with coefficient c by a finished term with coefficient $2c$ (its coefficients m_1, m' are prime to p) and an active term with coefficient $-c$ and $v_p = 1$;
- (ii) replaces it by an active term with coefficient $2c$ and $v_p = 1$, and a finished term with coefficient $-c$;
- (iii) replaces it by an active term with coefficient c and $v_p = 1$;
- (iv) replaces it by a finished term with coefficient c.

In every case $v_p(N_j)$ of the active term drops by at least one, the coefficients stay coprime (because $p \nmid m_1 m'$) and odd, and if $p \nmid N_j$ for the new active term we declare it finished. Since after any step the active term has $v_p(m_j m_j') = 1$, the steps (ii) and (iv) can only occur as the first step, and the recursion stops after $k \le v_p(N)$ steps. If the first step is (iv), the total is 1. If the first step is (ii), the active coefficient has absolute value 2 throughout, so the total of the absolute values of all coefficients is at most

$$\underbrace{1}_{\text{(ii)}} + \underbrace{4(k-1)}_{\text{later (i) steps}} + \underbrace{2}_{\text{last active term}} \le 4v_p(N) - 1.$$

Otherwise the active coefficient has absolute value 1 throughout, and the total is at most $2k + 1 \le 2v_p(N) + 1 \le 3v_p(N)$. In all cases the total is at most $4v_p(N)$.

Invariants (a), (b). Each of (i)–(iv) preserves $\tilde m \tilde m' \le mm'$ and does not increase $\tilde N/(\tilde m \tilde m')$: in (i) the products of the coefficients are $m_1 m' = mm'/p$ and $m_1 p m' = mm'$ while N becomes

N/p; in (ii) they are $pm_1m' = mm'/p$ (with N/p) and $m_1m' = mm'/p^2$ (with N/p^2); in (iii) $m_1pm' = mm'$ with N/p; in (iv) $m_1m' = mm'/p^2$ with N/p^2. They divide N only by powers of the odd prime p, so $\tilde{N} \mid N$ and $v_2(\tilde{N}) = v_2(N)$, and they do not change the p'-adic valuations of the coefficients or of $\tilde{N}$ for $p' \neq p$.

All primes. We now treat the primes of (mm', N) one after the other. By the last remark, the recursion at one prime neither creates nor destroys common factors at the other primes, so after all primes are treated every term satisfies $(\tilde{m}\tilde{m}', \tilde{N}) = 1$, and the total of the absolute values of the coefficients is at most $\prod_{p|(mm',N)} 4v_p(N)$. Since $v_p(N) < v_p(N) + 1$, this is $\leq 4^{\omega(N)} \prod_{p|N}(v_p(N) + 1) = 4^{\omega(N)} d(N)$.

Eisenstein and cuspidal parts. The operator U_p (p dividing the level) and the operator $\Phi(z) \mapsto \Phi(pz)$ map $M_2(\Gamma_0(L))$ to $M_2(\Gamma_0(pL))$, preserving both the subspace of cusp forms and the Eisenstein subspace. Since the decomposition $\Theta = E + G$ is unique, the identities hold separately for E and G, i.e. for M and for $T - M$. □

Remark 2.6 (Worked example). Take $p = 5$, $m' = 1$ and $N = 50$, so that $v_5(N) = 2$. For $m = 5$ the recursion is (i), followed by (i) with the roles of m, m' exchanged:

$$T(5, 1; 50) = 2T(1, 1; 10) - T(1, 5; 10), \qquad T(1, 5; 10) = 2T(1, 1; 2) - T(5, 1; 2),$$

so $T(5, 1; 50) = 2T(1, 1; 10) - 2T(1, 1; 2) + T(5, 1; 2)$; numerically, $244 = 2 \cdot 144 - 2 \cdot 24 + 4$. The three terms on the right have $5 \nmid \tilde{N}$, their coefficients are coprime and odd with $v_5 \leq 1$, $\tilde{N}/(\tilde{m}\tilde{m}') \in \{10, 2, 2/5\}$ is at most $N/(mm') = 10$, and the coefficient total $2 + 2 + 1 = 5$ equals $2v_5(N) + 1$. For $m = 25$ the first step is (ii), and the active term $T(5, 1; 10)$ it produces is finished by (i):

$$T(25, 1; 50) = 2T(5, 1; 10) - T(1, 1; 2), \qquad T(5, 1; 10) = 2T(1, 1; 2) - T(1, 5; 2),$$

so $T(25, 1; 50) = 3T(1, 1; 2) - 2T(1, 5; 2)$; numerically, $64 = 3 \cdot 24 - 2 \cdot 4$, with coefficient total $4 + 2 + 1 = 7 = 4v_5(N) - 1$ before the cancellation. If $v_5(N) = 1$, say $N = 10$, the recursion stops after one step: $T(5, 1; 10) = 2T(1, 1; 2) - T(1, 5; 2) = 44$ and $T(25, 1; 10) = 2T(5, 1; 2) - 0 = 8$. For an inert prime, $q = 3$ and $N = 18$: $T(3, 1; 18) = T(1, 3; 6) = T(3, 1; 2) = 4$ by (iii) applied twice, and $T(9, 1; 18) = T(1, 1; 2) = 24$ by (iv). All values were checked by direct enumeration of the representations.

Corollary 2.7. *For coprime odd m, m' with $v_p(mm') \leq 2$ for all p, and N with $v_2(N) \leq 1$,*

$$T(m, m'; N) = M(m, m'; N) + O_\varepsilon\big(4^{\omega(N)} d(N)^2 \, N^{1/2} (mm')^\varepsilon\big),$$

and $4^{\omega(N)} d(N)^2 \leq d(N)^4$. Moreover $M(m, m'; N) \ll_\eta 4^{\omega(N)} d(N) \, N^{1+\eta}/(mm')$ for every $\eta > 0$. (With Lemma 2.1 alone, the error term is $O_\varepsilon(4^{\omega(N)} d(N)^2 N^{1/2} (mm')^{1/4+\varepsilon})$.)

Proof. By Lemma 2.5, $T(m, m'; N) - M(m, m'; N) = \sum_j c_j \big(T(\tilde{m}_j, \tilde{m}'_j; \tilde{N}_j) - M(\tilde{m}_j, \tilde{m}'_j; \tilde{N}_j)\big)$ with $\sum_j |c_j| \leq 4^{\omega(N)} d(N)$, and each term satisfies the hypotheses of Theorem 2.3, which bounds it by

$$\ll d(\tilde{N}_j) \, \tilde{N}_j^{1/2} (\tilde{m}_j \tilde{m}'_j)^\varepsilon \leq d(N) \, N^{1/2} (mm')^\varepsilon,$$

because $\tilde{N}_j \mid N$ and $\tilde{m}_j \tilde{m}'_j \leq mm'$. This gives the first claim. Since $d(N) \geq 2^{\omega(N)}$ we have $4^{\omega(N)} \leq d(N)^2$, hence $4^{\omega(N)} d(N)^2 \leq d(N)^4$. For the bound on M apply (10) to each $M(\tilde{m}_j, \tilde{m}'_j; \tilde{N}_j)$ and use $\tilde{N}_j/(\tilde{m}_j \tilde{m}'_j) \leq N/(mm')$. □

2.5. **Proof of Theorem 1.2.** Throughout this subsection $Q = Q_{m,m'}$ has Gram matrix $A_Q = A = \operatorname{diag}(2m, 2m, 2m', 2m')$, $\det A_Q = 16(mm')^2$, level $L = 4mm'$, and $G = \Theta - E$. Nothing from Sections 2.3–2.4 is used. We use the following three facts from [18], in Waibel's notation ($m = 4$ variables, weight $m/2 = 2$; his N is our L, his Q our A_Q).

Lemma 2.8 ([18, (22)])**.** *Let $\mu = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(L)]$, $\tilde\mu = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma(L)]$, so that $\tilde\mu/\mu = L\varphi(L)$, let $\mathcal{F}$ be the standard fundamental domain of $\mathrm{SL}_2(\mathbb{Z})$ and $T = \{\left(\begin{smallmatrix}1 & x\\ 0 & 1\end{smallmatrix}\right) : \ 0 \le x \le L-1\}$. For a, c modulo L with $(a, c, L) = 1$ let $\varrho_{a,c} = \left(\begin{smallmatrix}a & *\\ c & *\end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. Then for $g \in S_2(\Gamma_0(L))$*

$$\|g\|^2 = \frac{\mu}{\tilde\mu} \sum_{c \ (L)} \sum_{\substack{a \ (L)\\ (a,c,L)=1}} \int_{\bigcup_{t\in T} t\mathcal{F}} \big|(g|_2\varrho_{a,c})(z)\big|^2 \,\mathrm{d}x\,\mathrm{d}y.$$

This is the fundamental domain of $\Gamma(L)$ written as a union of translates of $\mathcal{F}$ by coset representatives $\varrho_{a,c}t$ of $\Gamma(L)\backslash\mathrm{SL}_2(\mathbb{Z})$, together with $\langle g, g\rangle_{\Gamma_0(L)} = [\Gamma_0(L) : \Gamma(L)]^{-1}\langle g, g\rangle_{\Gamma(L)}$ and the invariance of $y^2|g|^2$.

Lemma 2.9 ([18, Lemma 12 and p. 15])**.** *Let $\varrho = \varrho_{a,c}$ with $c > 0$ and $d = (c, L) < L$, so that $\varrho \notin \Gamma_0(L)$. There are $\hat d \in \{d, d/2\}$, a diagonal integral matrix $D = D_\varrho$ and a positive integral quaternary form $S = S_\varrho$ of level at most L, depending on c only through d and the parity data of c, such that*

$$(G|_2\varrho)(z) = \sum_{n\ge 1} b_\varrho(n)\, e\Big(\frac{\hat d n z}{L}\Big), \qquad |b_\varrho(n)|^2 \le 32\,\frac{\det D}{\det A_Q}\big(\mathrm{r}(S,n)^2 + \mathrm{r}(\mathrm{gen}S,n)^2\big),$$

where $\mathrm{r}(S,n)$ is the number of representations of n by S and $\mathrm{r}(\mathrm{gen}S,n)$ the Siegel average over the genus of S. Explicitly, $S = (L/\hat d)\, D\tilde A^{-1}D$, where $\tilde A = U^{\mathrm{T}}A_QU$ with $U \in \mathrm{SL}_4(\mathbb{Z})$ chosen so that $\frac12 x^{\mathrm{T}}\tilde A x \equiv \sum_i q_i x_i^2$ modulo the odd part $\tilde c$ of c and is in Jordan form modulo $2^{v_2(c)}$, and $D = \operatorname{diag}(\tilde d_i 2^{t_i})$ with $\tilde d_i = (q_i, \tilde c, L)$ and $t_i \in \{0, 1\}$ determined by $v_2(c)$.

Here the factor $32 = 2\cdot 2^m$ accounts for the weights $\eta(x) \in \{0, 1, \sqrt2\}$ in Waibel's expansion and for $|u - v|^2 \le 2|u|^2 + 2|v|^2$; the Eisenstein part $E = \theta(\mathrm{gen}Q)$ transforms into the genus theta series of S, which is how $\mathrm{r}(\mathrm{gen}S, n)$ enters ([18, p. 15]).

Lemma 2.10 ([18, (14)])**.** *For a positive integral quaternary form S of level L_S,*

$$\mathrm{r}(\mathrm{gen}S, n) \ll_\varepsilon \frac{n\,(n, L_S)^{1/2}}{\sqrt{\det S}}\,(nL_S)^\varepsilon.$$

We now specialise Lemma 2.9 to $Q_{m,m'}$.

Lemma 2.11. *For $Q = Q_{m,m'}$ one can take $U = I$ in Lemma 2.9, and then, with $d_m = (m, c)$, $d_{m'} = (m', c)$ and $d_2 = (c, 4)$, so that $d = d_2d_md_{m'}$,*

$$D = \operatorname{diag}(d_m, d_m, d_{m'}, d_{m'})\cdot 2^t, \quad t \in \{0, 1\}, \qquad \tfrac12 x^{\mathrm{T}}Sx = A(x_1^2 + x_2^2) + B(x_3^2 + x_4^2),$$

where $A = \kappa\,(m'/d_{m'})\,d_m$ and $B = \kappa\,(m/d_m)\,d_{m'}$ are positive integers with $\kappa = 4^t d_m d_{m'}/\hat d \in \{\frac14, \frac12, 1, 2, 4, 8\}$. Put $g_0 = (d_m, m/d_m)\,(d_{m'}, m'/d_{m'})$, so that $g_0 = 1$ when m, m' are square-free, and in general $g_0 \mid d_md_{m'}$, hence $g_0 \le d \le 2\hat d$. Then

$$\det D \le 16\,d^2, \qquad AB = \kappa^2 mm' \ge \frac{mm'}{16}, \qquad \frac{(A, B)}{AB} \le \frac{4g_0}{mm'}, \qquad \sqrt{\det S} = 4AB \ge \frac{mm'}{4}.$$

Proof. The form $\frac12 x^{\mathrm{T}}A_Qx = m(x_1^2 + x_2^2) + m'(x_3^2 + x_4^2)$ is already diagonal, so $U = I$, $q = (m, m, m', m')$, there are no binary 2-adic blocks, and all 2-adic valuations $\nu_i(2)$ vanish; hence

$\tilde{d}_i = (q_i, \tilde{c}, L) = (q_i, c)$, which is d_m for $i = 1, 2$ and $d_{m'}$ for $i = 3, 4$, and the t_i are all equal to one value $t \in \{0, 1\}$. Then $A_Q^{-1} = \operatorname{diag}(1/(2m), 1/(2m), 1/(2m'), 1/(2m'))$ and

$$\tfrac{1}{2} x^{\mathrm{T}} S x = \frac{L}{\hat{d}} \cdot 4^t \Big(\frac{d_m^2 (x_1^2 + x_2^2)}{4m} + \frac{d_{m'}^2 (x_3^2 + x_4^2)}{4m'} \Big) = \frac{4^t}{\hat{d}} \big(m' d_m^2 (x_1^2 + x_2^2) + m d_{m'}^2 (x_3^2 + x_4^2) \big),$$

which is the stated form with $\kappa = 4^t d_m d_{m'} / \hat{d} = 4^t / \hat{d}_2$, $\hat{d}_2 = \hat{d}/(d_m d_{m'}) \in \{d_2, d_2/2\} \subset \{\frac{1}{2}, 1, 2, 4\}$. The integrality of S (Waibel's Lemma 11) forces $A, B \in \mathbb{Z}$. Write $A = \kappa a$, $B = \kappa b$ with $a = (m'/d_{m'}) d_m$, $b = (m/d_m) d_{m'}$. Since $(m, m') = 1$, the primes of d_m and m/d_m divide m and those of $d_{m'}$ and $m'/d_{m'}$ divide m', so $(a, b) = (d_m, m/d_m)(d_{m'}, m'/d_{m'}) = g_0$; note that $g_0 > 1$ is possible when m or m' is not square-free (e.g. $p^2 \mid m$ and $p \,\|\, c$ give $d_m = p = m/d_m$ up to units, so $p \mid g_0$). If $\kappa \in \mathbb{Z}$ then $(A, B) = \kappa g_0$; if $\kappa \in \{\frac{1}{2}, \frac{1}{4}\}$ then $A, B \in \mathbb{Z}$ forces $\kappa^{-1} \mid a, b$ and $(A, B) = \kappa g_0$ as well. Hence $(A, B)/(AB) = \kappa g_0 / (\kappa^2 ab) = g_0/(\kappa m m') \le 4 g_0/(m m')$. Finally $g_0 \mid d_m d_{m'} \mid d$ and $d \le 2\hat{d}$, $\det D = (d_m d_{m'})^2 2^{4t} \le 16 d^2$ and $\det S = \det \operatorname{diag}(2A, 2A, 2B, 2B) = 16 A^2 B^2$. □

Lemma 2.12. *Let $A, B \ge 1$ be integers with $(A, B)/(AB) \le 4 g_0/(m m')$ for some $g_0 \ge 1$. Then for every $n \ge 1$,*

$$\mathrm{r}\big(A(x_1^2 + x_2^2) + B(x_3^2 + x_4^2),\, n\big) \;\ll_\varepsilon\; n^\varepsilon \Big(1 + \frac{g_0\, n}{m m'} \Big).$$

Proof. A representation gives $\alpha = x_1^2 + x_2^2$, $\beta = x_3^2 + x_4^2$ with $A\alpha + B\beta = n$, $\alpha, \beta \ge 0$. Put $g = (A, B)$; there are no solutions unless $g \mid n$, and then the solutions α lie in one residue class modulo B/g and in $[0, n/A]$, so there are at most $ng/(AB) + 1 \le 4 g_0 n/(m m') + 1$ pairs (α, β), and each pair contributes $r(\alpha) r(\beta) \le 16\, d(\alpha) d(\beta) \ll n^\varepsilon$ representations. □

Proof of Theorem 1.2. Apply Lemma 2.8 to $g = G$. Fix $\varrho = \varrho_{a,c}$ and put $I(\varrho) = \int_{\cup_t t\mathcal{F}} |G|_2 \varrho|^2 \, \mathrm{d}x \, \mathrm{d}y$. The region $\cup_{t \in T} t\mathcal{F}$ lies in the strip $\{-\frac{1}{2} \le x \le L - \frac{1}{2},\ y \ge \sqrt{3}/2\}$, and $G|_2 \varrho$ is periodic in x with a period dividing L, being a form on $\varrho^{-1} \Gamma_0(L) \varrho \supseteq \Gamma(L)$. Hence, if $G|_2 \varrho = \sum_n b(n) e(\hat{d} n z / L)$, Parseval over a full period gives

$$I(\varrho) \le L \sum_{n \ge 1} |b(n)|^2 \int_{\sqrt{3}/2}^{\infty} e^{-4\pi \hat{d} n y / L} \, \mathrm{d}y = \frac{L^2}{4\pi \hat{d}} \sum_{n \ge 1} \frac{|b(n)|^2}{n} \, e^{-2\sqrt{3}\pi \hat{d} n / L}. \tag{14}$$

The cusp ∞. If $d = L$, then $\varrho \in \Gamma_0(L)$, $b(n) = a_G(n)$ and $\hat{d} = L$. For every n, $|a_G(n)| \le T(m, m'; n) + M(m, m'; n) \ll n^\varepsilon (1 + n/(m m'))$ by Lemma 2.12 with $A = m$, $B = m'$, $g_0 = 1$ (for T) and Lemma 2.10 applied to Q itself (for M; here $(n, L)^{1/2} \le n^{1/2}$ and $n^{3/2}/(m m') \le n^{3/2}$ is harmless because of the exponential). Hence $I(\varrho) \ll L \sum_n n^{2\varepsilon} (1 + n^{3/2}) e^{-2\sqrt{3}\pi n} \ll L$, and these $\varphi(L)$ pairs (a, c) contribute $\ll (\mu/\tilde{\mu}) \varphi(L) L = 1$ to the sum in Lemma 2.8.

The other cusps. Let $d \mid L$, $d < L$. The number of c modulo L with $(c, L) = d$ is $\varphi(L/d)$, and for each such c the number of a modulo L with $(a, c, L) = 1$ is $\varphi(d) L/d$; since $\varphi(L/d)\varphi(d) \le \varphi(L)$, these pairs contribute at most

$$\frac{\mu}{\tilde{\mu}} \varphi(L/d)\, \varphi(d)\, \frac{L}{d} \max_{\varrho} I(\varrho) \;\le\; \frac{1}{d} \max_{\varrho} I(\varrho),$$

the maximum over the $\varrho_{a,c}$ with $(c, L) = d$. By Lemma 2.9, Lemma 2.11 and (14), with $L^2 = 16 (m m')^2$, $\det A_Q = 16 (m m')^2$, $\det D \le 16 d^2$ and $d/\hat{d} \le 2$,

$$\begin{aligned} \frac{1}{d} I(\varrho) &\le \frac{L^2}{4\pi d \hat{d}} \cdot \frac{32 \cdot 16 d^2}{16 (m m')^2} \sum_n \frac{\mathrm{r}(S, n)^2 + \mathrm{r}(\mathrm{gen} S, n)^2}{n} e^{-2\sqrt{3}\pi \hat{d} n / L} \\ &\le \frac{256}{\pi} \sum_n \frac{\mathrm{r}(S, n)^2 + \mathrm{r}(\mathrm{gen} S, n)^2}{n} e^{-2\sqrt{3}\pi \hat{d} n / L}. \end{aligned}$$

For the first sum, Lemmas 2.11 and 2.12 give $\mathrm{r}(S,n)^2 \ll n^{2\varepsilon}(1+g_0 n/(mm'))^2$ with $g_0 \le 2\hat{d}$. We keep $\hat{d}$ in the exponential and split at $n = 2L/\hat{d}$: for $n \le 2L/\hat{d}$ we have $g_0 n/(mm') \le 4L/(mm') = 16$, and for the tail we use $\sum_{n>Z} n\, e^{-\lambda n} \ll \lambda^{-2}$ with $\lambda = 2\sqrt{3}\pi\hat{d}/L$. Hence

$$\sum_n \frac{n^{2\varepsilon}(1+g_0 n/(mm'))^2}{n} e^{-2\sqrt{3}\pi\hat{d}n/L} \ll 17^2 \sum_{n\le 2L/\hat{d}} n^{2\varepsilon-1} + \frac{L^{2\varepsilon} g_0^2}{(mm')^2} \sum_{n>2L/\hat{d}} n\, e^{-2\sqrt{3}\pi\hat{d}n/L}$$
$$\ll L^{2\varepsilon}\log L + L^{2\varepsilon}\,\frac{g_0^2}{\hat{d}^2}\cdot\frac{L^2}{(mm')^2} \ll L^{3\varepsilon},$$

since $g_0/\hat{d} \le 2$ and $L = 4mm'$, so that the last fraction is at most 64. (For square-free m, m' one has $g_0 = 1$ and the splitting is not needed; the care with $\hat{d}$ matters only when m or m' has a square factor, which does occur in Section 3, where the coefficients are cube-free.) For the second sum, Lemma 2.10 with $L_S \le L$ and $\sqrt{\det S} \ge mm'/4$ gives $\mathrm{r}(\mathrm{gen}S,n)^2 \ll L^{2\varepsilon} n^2 (n,L)/(mm')^2$, and

$$\sum_n \frac{n^2(n,L)}{(mm')^2 n} e^{-2\sqrt{3}\pi\hat{d}n/L} \le \frac{1}{(mm')^2}\sum_{g\mid L} g \sum_{k\ge 1} gk\, e^{-\sqrt{3}\pi g k/L}$$
$$\ll \frac{1}{(mm')^2}\sum_{g\mid L} g^2\Big(\frac{L}{g}\Big)^2 = \frac{d(L)L^2}{(mm')^2} \ll L^{\varepsilon},$$

where we wrote $n = gk$ with $g = (n,L)$ and used $\sum_k k\, e^{-\lambda k} \ll \lambda^{-2}$. Hence each class $d < L$ contributes $\ll L^{3\varepsilon}$, there are $d(L) \ll L^{\varepsilon}$ classes, and

$$\|G\|^2 \ll 1 + d(L) L^{3\varepsilon} \ll L^{4\varepsilon}.$$

Renaming ε gives the theorem. □

Remark 2.13. The proof shows more precisely that

$$\|G_{m,m'}\|^2 \ll d(mm')\,(\log mm')^2 \max_{n\le 16mm'} \max_{\alpha,\beta\le n} \big(d(\alpha)d(\beta)\big)^2,$$

so that the true order of the norm is governed by the size of $r(\alpha)r(\beta)$ on the $O(1)$ pairs (α,β) with $A\alpha + B\beta = n$, $n \ll mm'$. On average over n these products are of size $\log n$, which suggests $\|G_{m,m'}\|^2 \asymp (\log mm')^2$; the data of Part II [4, Section 5] give $\|G_{m,m'}\|^2 \approx 0.37(\log mm')^2$ for $10^2 \le mm' \le 1.6\cdot 10^4$. Part II [4, Theorem 1.5] proves that $\|G_{m,m'}\| = \|G_{1,mm'}\|$ (the Atkin–Lehner involution W_m maps $G_{m,m'}$ to $\pm G_{1,mm'}$) and the lower bound $\|G_{m,m'}\|^2 \ge \frac{1}{\pi}(\log mm')^2 - O\big(\log(2mm')(\log\log 3mm')^C\big)$, by integrating $|G_{1,mm'}|^2$ over the parts of the fundamental domain above the cusps $W_{mm'}\infty$ and $W_{4mm'}\infty$, where the first mm' coefficients are $r(n) - M(1,mm';n)$; so the exponent ε in Theorem 1.2 cannot be replaced by 0; and Part II, Theorem 6.5, proves the matching upper bound $\|G_{m,m'}\|^2 \ll (\log 3mm')^2(\log\log 3mm')^C$ by replacing the two ε-losses of the proof above (the bound $r(\alpha)r(\beta) \le 16 d(\alpha)d(\beta)$ in Lemma 2.12 and Lemma 2.10) by mean values, so that the norm is of the exact order $(\log mm')^2$ when mm' has boundedly many prime factors, and Theorem 2.3 holds with $(\log 3mm')(\log\log 3mm')^C$ in place of $(mm')^{\varepsilon}$. Numerically the regions above all the cusps carry 93–98% of the norm (Part II, Section 6). Waibel's Theorem 1 gives $(mm')^{1/2+\varepsilon}$ for $\|G\|^2$, and his Theorem 2 for diagonal forms gives $\min(m,m')(mm')^{\varepsilon}$; both are sharp for other families and both lose here because they bound $\mathrm{r}(S,n)$ by the general lattice-point estimate of [18, Lemma 9].

3. The level of distribution of $r(n)r(N-n)$

Let $N \equiv 2 \pmod 4$ and

$$\mathcal{A} = \{n : \ 1 \le n \le N, \ (n, N) = 1\}, \qquad w(n) = r(n)r(N-n).$$

Elements of $\mathcal{A}$ are odd, and $w(n) \neq 0$ forces $n \equiv N - n \equiv 1 \pmod 4$. For coprime square-free d, e with $(de, 2N) = 1$ put

$$S(d, e) = \sum_{\substack{n \in \mathcal{A} \\ d|n, \ e|N-n}} w(n).$$

3.1. **Decomposition into theta coefficients.** For split p and inert q the relations $r(n) = 2r(n/p) - r(n/p^2)$ $(p \mid n)$ and $r(n) = r(n/q^2)$ $(q \mid n)$ give, for square-free d,

$$\sum_{d|n} r(n)e(nz) = \sum_{m|d^2} c_d(m)\, \theta^2(mz), \tag{15}$$

where c_d is multiplicative in the sense $c_d(m) = \prod_{p|d} c_p(p^{v_p(m)})$ with $c_p(p) = 2$, $c_p(p^2) = -1$ for split p and $c_q(q^2) = 1$, $c_q(q) = 0$ for inert q. Every m with $c_d(m) \neq 0$ has the form $m = ds$ with $s \mid d$, and $\sum_m |c_d(m)| \le 3^{\omega(d)}$. Moreover every such m is $\equiv 1 \pmod 4$, being a product of split primes and of squares of inert primes. The condition $(n, N) = 1$ is removed by Möbius inversion over the square-free divisors t of the odd part of N, which brings in the identity (15) once more with t in place of d, and the parity of n, equivalently of the variable a in (2), by the identity $r(2a') = r(a')$: the terms with a even in $T(m, m'; N)$ sum to $T(m, m'; N/2)$. Altogether

$$S(d, e) = \sum_{t|N^\flat} \mu(t) \sum_{m|d^2} \sum_{m'|e^2} c_d(m)c_e(m') \sum_{t_1|t^2} c_t(t_1)\, T^{\mathrm{odd}}(t_1 m, m'; N), \tag{16}$$

with $T^{\mathrm{odd}}(\cdot\,; N) = T(\cdot\,; N) - T(\cdot\,; N/2)$, where $N^\flat$ is the product of the odd primes dividing N. Each T in (16) has coprime, odd, cube-free coefficients $t_1 m \equiv m' \equiv 1 \pmod 4$ and is covered by Corollary 2.7; by (8), the congruence ensures that all the forms $Q_{t_1 m, m'}$ occurring have the same 2-adic densities as $Q_{1,1}$.

3.2. **The main term.** Define $M(d, e)$ by replacing each T in (16) by its Eisenstein part M. The Siegel densities are multiplicative over the primes, and the linear identities behind (15) and the Möbius inversion hold for the numbers of solutions modulo every power of an odd prime; at $p = 2$ all the forms in (16) have the densities of $Q_{1,1}$ by (8), and the parity condition selects the 2-adic solutions of $x^2 + y^2 + z^2 + w^2 = N$ with $x^2 + y^2$ odd, whose density is $\beta_2(N) - \frac{1}{2}\beta_2(N/2) = 1$ independently of d, e. Hence the ratios $M(d, e)/X$ below are products of conditional local densities at the odd primes. Concretely, let X be the Eisenstein part of $S(1, 1) = \sum_{n \in \mathcal{A}} w(n)$:

$$X = \big(r_4(N) - r_4(N/2)\big) \prod_{p|N^\flat} (1 - g_N(p)) = 16\, \sigma(N/2) \prod_{p|N^\flat} (1 - g_N(p)), \tag{17}$$

where we used that $M(1, 1; n) = r_4(n)$, Jacobi's formula ($r_4(N) = 24\sigma(N/2)$, $r_4(N/2) = 8\sigma(N/2)$ for $N \equiv 2 \pmod 4$), and where $g_N(p)$ is the proportion of the p-adic solutions of $x^2+y^2+z^2+w^2 = N$ with $p \mid x^2 + y^2$. Then

$$M(d, e) = X\, h(d)\, h(e), \qquad h(p) = \frac{2p - 1}{p(p + 1)} \ \ (p \equiv 1 \ (4)), \qquad h(q) = \frac{1}{q(q - 1)} \ \ (q \equiv 3 \ (4)), \tag{18}$$

h multiplicative.

Lemma 3.1. *Let p be odd. (i) If $p \nmid N$, the proportion of the p-adic solutions of $x^2+y^2+z^2+w^2 = N$ with $p \mid x^2+y^2$ is $h(p)$. (ii) If $p^a \parallel N$, $a \ge 1$, this proportion is*

$$g_N(p) = 1 - \frac{\beta_p^*}{(1-p^{-2})\sigma_{-1}(p^a)}, \qquad \beta_p^* = \begin{cases} (1-1/p)^3 & p \equiv 1 \pmod 4, \\ (1-1/p)(1+1/p)^2 & p \equiv 3 \pmod 4; \end{cases}$$

in particular $g_N(p) = 4p/(p+1)^2$ for $p \equiv 1 \pmod 4$, $p \parallel N$, and $g_N(q) = 0$ for $q \equiv 3 \pmod 4$, $q \parallel N$.

Proof. (i) Modulo p, with N a unit, the equation $x^2+y^2+z^2+w^2 \equiv N$ has p^3-p solutions, all non-singular. Those with $x^2+y^2 \equiv 0$ number $(2p-1)(p-1)$ if $p \equiv 1 \pmod 4$ ($2p-1$ choices of (x,y) on the two lines $y \equiv \pm ix$, then $p-1$ choices of (z,w) with $z^2+w^2 \equiv N$) and $1 \cdot (q+1)$ if $q \equiv 3 \pmod 4$. Hensel's lemma lifts each solution to $p^{3(a-1)}$ solutions modulo p^a, so the proportion is $(2p-1)(p-1)/(p^3-p) = h(p)$, resp. $(q+1)/(q^3-q) = h(q)$. (ii) The solutions with $p \nmid x^2+y^2$ are non-singular, and their number modulo p is $\sum_{u \not\equiv 0} \#\{x^2+y^2 \equiv u\}\#\{z^2+w^2 \equiv -u\} = (p-1)(p-\chi(p))^2$, which gives the density β_p^*; the total density is (6). □

Note that $(1-g_N(p))\sigma_{-1}(p^a) = \beta_p^*/(1-p^{-2})$ does not depend on a; this is why $\mathfrak{S}_r(N)$ in (1) has one factor per prime. Note also that for $p \nmid N$ the events $p \mid n$ and $p \mid N-n$ are mutually exclusive on $\mathcal{A}$, each of density $h(p)$, so the density of $\{n \in \mathcal{A} :\ p \nmid n(N-n)\}$ is $1-2h(p)$, and a direct computation gives

$$\frac{\beta_p^*}{(1-p^{-2})(1-2h(p))} = \frac{p-1}{p-2} \qquad \text{for split and inert } p \text{ alike,} \tag{19}$$

since $1-2h(p) = (p-1)(p-2)/(p(p+1))$ for $p \equiv 1 \pmod 4$ and $1-2h(q) = (q-2)(q+1)/(q(q-1))$ for $q \equiv 3 \pmod 4$. This is the local factor in (1). The products in (17) and (18) follow from Lemma 3.1 by multiplicativity; for (18) one can also verify directly that $\sum_{m \mid d^2} c_d(m)\, g(\operatorname{rad} m)/m = h(d)$ with g as in (9): for split p, $2g(p)/p - g(p)/p^2 = \frac{p}{p+1}\left(\frac{2}{p} - \frac{1}{p^2}\right) = h(p)$, and for inert q, $g(q)/q^2 = h(q)$.

3.3. **The remainder.** Put $R(d,e) = S(d,e) - M(d,e)$.

Theorem 3.2. *Let $N \equiv 2 \pmod 4$, $\varepsilon > 0$ and $D = N^{1/2-\varepsilon}$. Then*

$$\sum_{\substack{de \le D,\ (d,e)=1 \\ (de,2N)=1}} \mu^2(de)\, d_3(de)\, |R(d,e)| \ \ll_\varepsilon \ N^{1-\varepsilon/3}.$$

The same holds with $D = N^{2/5-\varepsilon}$ if Corollary 2.7 is used with the error term $(mm')^{1/4+\varepsilon}$ of Lemma 2.1.

Proof. Set-up. By (16), $R(d,e)$ is a sum of terms $\pm c\,(T-M)(\tilde m, \tilde m'; N')$ with $N' \in \{N, N/2\}$ and coefficients $(\tilde m, \tilde m') = (t_1 m, m')$, $m = ds$, $m' = es'$, $s \mid d$, $s' \mid e$, $t_1 \mid t^2$, $t \mid N^\flat$; thus

$$\tilde m \tilde m' = t_1 s s' \cdot de.$$

These coefficients are coprime, odd and cube-free because d, e, t are pairwise coprime and square-free, so Corollary 2.7 applies (with $v_2(N') \le 1$). Let $\eta = \varepsilon/10$. We use the following facts, each of which holds for $N \ge N_0(\varepsilon)$:

- (F1) The product of the divisor-type factors attached to one term, namely $d_3(de)$, the coefficient $|c_d(m)c_e(m')c_t(t_1)| \le 3^{\omega(d)+\omega(e)+\omega(t)}$, and the factor $4^{\omega(N)} d(N)^2 \le d(N)^4$ of Corollary 2.7, is $\le N^\eta$.
- (F2) For fixed (d,e), the number of tuples (s,s',t,t_1) is at most $d(d)d(e)\, d(N^\flat)\, 3^{\omega(N)} \le N^\eta$.
- (F3) $\sum_{k \le D} d(k)/k \le N^\eta$ and $\sum_{k \le D} d(k) \le D N^\eta$; here $d(k)$ counts the pairs (d,e) with $de = k$.

(F4) $\tilde{m}\tilde{m}' = t_1 ss' \cdot de \le N^{\varepsilon} D = N^{1/2} \le N$ whenever $t_1 ss' \le N^{\varepsilon}$, so $(\tilde{m}\tilde{m}')^{\eta} \le N^{\eta}$.

We use two bounds for $|T - M|$. The first is Corollary 2.7:

$$|T(\tilde{m}, \tilde{m}'; N') - M(\tilde{m}, \tilde{m}'; N')| \ll 4^{\omega(N)} d(N)^2 N^{1/2} (\tilde{m}\tilde{m}')^{\eta}. \tag{20}$$

The second is the trivial bound

$$|T - M| \le T + M \ll N^{2\eta}\Big(\frac{N}{\tilde{m}\tilde{m}'} + 1\Big) : \tag{21}$$

for T, the pairs (a, b) with $\tilde{m}a + \tilde{m}'b = N'$ have a in a fixed residue class modulo $\tilde{m}'$ (as $(\tilde{m}, \tilde{m}') = 1$) and $0 \le a \le N/\tilde{m}$, so there are at most $N/(\tilde{m}\tilde{m}') + 1$ of them, and $r(a)r(b) \ll N^{\eta}$; for M we use the last assertion of Corollary 2.7. We split the terms according to the size of $t_1 ss'$.

Terms with $t_1 ss' > N^{\varepsilon}$. By (21), (F1) and (F2), their total contribution is at most

$$\ll N^{\eta} \cdot N^{\eta} \sum_{de \le D} \Big(\frac{N^{1+2\eta-\varepsilon}}{de} + N^{2\eta}\Big) \ll N^{1-\varepsilon+4\eta} \sum_{k \le D} \frac{d(k)}{k} + N^{4\eta} \sum_{k \le D} d(k) \ll N^{1-\varepsilon+5\eta} + N^{1/2+5\eta}.$$

Terms with $t_1 ss' \le N^{\varepsilon}$. By (20), (F1), (F2) and (F4), $(\tilde{m}\tilde{m}')^{\eta} \le N^{\eta}$, and their total contribution is at most

$$\ll N^{\eta} \cdot N^{\eta} \cdot N^{1/2} \cdot N^{\eta} \sum_{k \le D} d(k) \ll N^{1/2+4\eta} D = N^{1-\varepsilon+4\eta}.$$

Conclusion. Both contributions are $\ll N^{1-\varepsilon+5\eta} + N^{1/2+5\eta}$. With $\eta = \varepsilon/10$ this is $N^{1-\varepsilon/2} + N^{1/2+\varepsilon/2}$, and both are $\le N^{1-\varepsilon/3}$ for $\varepsilon \le 1/2$ (the second because $1/2 + \varepsilon/2 \le 1 - \varepsilon/3$ for $\varepsilon \le 3/5$). For $\varepsilon > 1/2$ we have $D < 1$ and the sum is empty.

The level 2/5 from Lemma 2.1. If (20) is replaced by the bound $4^{\omega(N)} d(N)^2 N^{1/2} (\tilde{m}\tilde{m}')^{1/4+\eta}$ of Corollary 2.7 with Lemma 2.1, take $D = N^{2/5-\varepsilon}$. For $t_1 ss' \le N^{\varepsilon}$ we have $(\tilde{m}\tilde{m}')^{1/4+\eta} \le N^{\eta+\varepsilon/4} (de)^{1/4}$, and with $\sum_{k \le D} d(k) k^{1/4} \le D^{5/4} N^{\eta}$ the contribution is $\ll N^{1/2+\varepsilon/4+4\eta} D^{5/4} = N^{1-\varepsilon+4\eta}$, since $D^{5/4} = N^{1/2-5\varepsilon/4}$; the other terms contribute $\ll N^{1-\varepsilon+5\eta} + N^{2/5+5\eta}$ as before, and the conclusion is the same. □

Remark 3.3. The exponent $1/2$ is $1/(2(1+\beta))$ with $\beta = 0$ the exponent of mm' in Theorem 2.3: the balancing condition is $N^{1/2} D^{1+\beta} \cdot D \approx N$ with the factor D^{β} from the level $\tilde{m}\tilde{m}' \approx de \le D$ and the factor D from the number of moduli. The terms with m close to d^2, where the level $\tilde{m}\tilde{m}'$ is as large as $(de)^2$, are harmless because of the trivial bound. With Waibel's bound alone, $\beta = 1/4$ and the level is $2/5$; the bound $|a_G(N)| \ll \|G\| N^{1/2+\varepsilon}(1 + N^{1/4}(mm')^{-1/2})$ obtained from the Petersson formula directly (Section 2), with $\|G\| \ll (mm')^{1/4+\varepsilon}$, has in the relevant range $mm' \le N^{1/2}$ the shape $N^{3/4+\varepsilon}(mm')^{-1/4}$, and the same balancing gives $N^{3/4} D^{3/4} \approx N$, i.e. the level $1/3$. For reference, the exponents of the proof are collected in the following ledger ($\eta = \varepsilon/10$):

	terms with $t_1 ss' \le N^{\varepsilon}$	terms with $t_1 ss' > N^{\varepsilon}$
one $\lvert T - M \rvert$	$N^{1/2}(\tilde{m}\tilde{m}')^{\eta}$ by (20)	$N^{2\eta}(N/(\tilde{m}\tilde{m}') + 1)$ by (21)
level $\tilde{m}\tilde{m}' = t_1 ss' de$	$(\tilde{m}\tilde{m}')^{\eta} \le N^{\eta}$	$N/(\tilde{m}\tilde{m}') < N^{1-\varepsilon}/(de)$
divisor factors, (F1) and (F2)	$N^{2\eta}$	$N^{2\eta}$
sum over $de = k \le D$, (F3)	$\sum d(k) \le DN^{\eta}$	$\sum d(k)/k \le N^{\eta}$, $\sum d(k) \le DN^{\eta}$
total	$N^{1/2+4\eta} D = N^{1-\varepsilon+4\eta}$	$N^{1-\varepsilon+5\eta} + N^{1/2+5\eta}$

With Waibel's bound ($\beta = 1/4$, $D = N^{2/5-\varepsilon}$) the left column reads $N^{1/2}(\tilde{m}\tilde{m}')^{1/4+\eta}$, $(\tilde{m}\tilde{m}')^{1/4+\eta} \le N^{\varepsilon/4+\eta}(de)^{1/4}$, $N^{2\eta}$, $\sum d(k) k^{1/4} \le D^{5/4} N^{\eta}$ and $N^{1/2+\varepsilon/4+4\eta} D^{5/4} = N^{1-\varepsilon+4\eta}$, and the right column acquires the term $N^{2/5+5\eta}$ in place of $N^{1/2+5\eta}$.

4. The sieve

We keep the notation of Section 3. Let $0 < \theta \le 1/2$ be a level of distribution for which Theorem 3.2 holds with $D = N^{\theta-\varepsilon}$ (by Theorem 3.2, $\theta = 1/2$, and hence every smaller θ, is admissible; $\theta = 2/5$ is admissible on the basis of Lemma 2.1 alone). Fix $0 < \alpha < \beta_1 < \theta$ and $\lambda > 0$, and put

$$z = N^{\alpha}, \qquad y_1 = N^{\beta_1}, \qquad P = \prod_{\substack{2<p<z \\ p\nmid N}} p.$$

For $n \in \mathcal{A}$ define Richert's weight

$$w_n = 1 - \lambda \sum_{\substack{p \mid n(N-n),\ p\equiv 1\,(4) \\ z \le p < y_1}} \Big(1 - \frac{\log p}{\log y_1}\Big), \qquad W = \sum_{\substack{n\in\mathcal{A} \\ (n(N-n),P)=1}} w_n\, w(n).$$

Lemma 4.1 (Structure)**.** *Let $n \in \mathcal{A}$ with $(n(N-n), P) = 1$ and $w(n) \neq 0$. Then every prime factor of $n(N-n)$ is $\ge z$. Unless $n(N-n)$ is divisible by the square of a prime $\ge z$, both n and $N-n$ are products of distinct primes $\equiv 1 \pmod 4$, each prime dividing exactly one of them, and $w(n) = 16 \cdot 2^{\Omega(n(N-n))}$. The total weight of the exceptional n is $\ll N^{1-\alpha+\varepsilon}$.*

Proof. $(n, N) = 1$ and $(n(N-n), P) = 1$ leave only prime factors $\ge z$. An inert prime divides a sum of two squares only to an even power, so if $n(N-n)$ is square-free all its prime factors are split, and then $r(n) = 4 \cdot 2^{\omega(n)}$. For the exceptional n, $p^2 \mid n(N-n)$ for some $p \ge z$, and $\sum_{p\ge z} \sum_{p^2 \mid n(N-n)} w(n) \ll N^{\varepsilon} \sum_{p\ge z} N/p^2 \ll N^{1-\alpha+\varepsilon}$. □

Lemma 4.2 (Prime factors)**.** *If n is as in Lemma 4.1, $n(N-n)$ square-free and $w_n > 0$, then $\Omega(n(N-n)) < 1/\lambda + 2/\beta_1$ and $\max(\Omega(n), \Omega(N-n)) < 1/\lambda + 1/\beta_1$.*

Proof. As in [3, Lemma 3.2]. Write $w_n = 1 - \lambda(T_1 + T_2)$ with T_1, T_2 the sums over the primes dividing n, resp. $N-n$. Since n is square-free with all prime factors $\ge z$, $\Omega(n) = \omega(n)$ and every prime factor of n below y_1 contributes to T_1, so

$$T_1 = \#\{p \mid n:\ p < y_1\} - \sum_{\substack{p\mid n \\ p<y_1}} \frac{\log p}{\log y_1} \ \ge\ \#\{p \mid n:\ p < y_1\} - \frac{\log n}{\log y_1} + \sum_{\substack{p\mid n \\ p\ge y_1}} \frac{\log p}{\log y_1} \ >\ \Omega(n) - \frac{1}{\beta_1},$$

because $\sum_{p\mid n} \log p = \log n < \log N = \beta_1^{-1} \log y_1$ and each term of the last sum is ≥ 1. Similarly $T_2 > \Omega(N-n) - 1/\beta_1$. All terms of T_1, T_2 are non-negative, so $w_n > 0$ gives $T_1 + T_2 < 1/\lambda$; hence $\Omega(n(N-n)) < 1/\lambda + 2/\beta_1$, and $\Omega(n) < T_1 + 1/\beta_1 \le T_1 + T_2 + 1/\beta_1 < 1/\lambda + 1/\beta_1$, likewise for $N-n$. □

Proposition 4.3 (Main term)**.** *Let $\theta_A \in (0, \theta)$, $s_A = \theta_A/\alpha$, $s_B = (\theta - \theta_A)/\alpha$, and let F_1, f_1 be the upper and lower functions of the linear sieve. Put*

$$L = f_1(s_A)F_1(s_B) + F_1(s_A)f_1(s_B) - F_1(s_A)F_1(s_B), \qquad R = 2\int_{\alpha}^{\beta_1} \Big(1 - \frac{u}{\beta_1}\Big) U(\theta - u)\, \frac{\mathrm{d}u}{u},$$

where $U(t) = \min_{t_A + t_B = t} F_1(t_A/\alpha) F_1(t_B/\alpha)$. Then

$$W \ \ge\ (L - \lambda R + o(1))\, X\, V, \qquad V = \prod_{\substack{2<p<z \\ p\nmid N}} (1 - 2h(p)),$$

and

(22)

$$XV = \big(1 + o(1)\big)\, 8\, \mathfrak{S}_r(N)\, c_{2h}\, \frac{N}{\alpha^2 (\log N)^2}, \qquad c_{2h} = \lim_{z\to\infty} (\log z)^2 \prod_{2<p<z} (1 - 2h(p)) = 1.66486\ldots$$

Proof. Densities. By (18), for coprime square-free d, e with $(de, 2N) = 1$ the subsequence $\{n \in \mathcal{A} : \ d \mid n,\ e \mid N-n\}$ with the weight w has density $h(d)h(e)$, where $h(p) = (2p-1)/(p(p+1))$ on the split primes and $h(q) \ll q^{-2}$ on the inert ones. If $(d, e) > 1$ then $S(d, e) = 0$, since a prime dividing d, e and hence n, $N - n$ would divide N. So for every $p \nmid 2N$ the events $p \mid n$ and $p \mid N-n$ are mutually exclusive and each has density $h(p)$: the density of $\{n \in \mathcal{A} : \ p \nmid n(N-n)\}$ is $1 - 2h(p)$, not $(1 - h(p))^2$. A vector sieve applied to the two components with the product density would therefore misstate the main term by the constant factor $\prod_p (1-2h(p))/(1-h(p))^2$. We avoid this by sifting the small primes from the product $n(N-n)$, where the density $\prod_{p \mid d_0} 2h(p)$ is exact, and only the large primes by the vector sieve, where the discrepancy at a prime p is a factor $1 + O(p^{-2})$. Since $\sum_{p<z,\, p\equiv 1\,(4)} 1/p = \frac{1}{2}\log\log z + O(1)$ and $h(p) = 2/p + O(p^{-2})$ on these primes, the function h satisfies the two-sided linear condition

$$\prod_{u \le p < v,\ p \nmid N} (1 - h(p))^{-1} = \frac{\log v}{\log u}\Big(1 + O\Big(\frac{1}{\log u}\Big)\Big) \qquad (2 \le u < v), \tag{23}$$

by the prime number theorem in progressions, and $2h$ satisfies the analogous condition of dimension 2.

Set-up. Let $z_0 = (\log N)^5$, $s_0 = \log\log N$, $D_0 = z_0^{s_0} = N^{o(1)}$, and split $P = P_0 P_1$ with $P_0 = \prod_{2<p<z_0,\, p \nmid N} p$. By the fundamental lemma of the β-sieve of dimension 2 [6, Chapter 6] there are weights $\rho_{d_0}^{\pm}$ supported on $d_0 \mid P_0$, $d_0 < D_0$, with $|\rho_{d_0}^{\pm}| \le 1$,

$$\sum_{d_0 \mid m} \rho_{d_0}^{-} \le 1_{(m, P_0)=1} \le \sum_{d_0 \mid m} \rho_{d_0}^{+} \quad (m \ge 1), \qquad \sum_{d_0} \rho_{d_0}^{\pm}\, \omega_0(d_0) = V_0\,(1 + o(1)),$$

where $\omega_0(d_0) = \prod_{p \mid d_0} 2h(p)$ and $V_0 = \prod_{p \mid P_0}(1 - 2h(p))$; the $o(1)$ tends to 0 as $s_0 \to \infty$, uniformly for densities satisfying the dimension-2 condition with a fixed constant. Note that $\omega_0(d_0)$ is the density of $\{n \in \mathcal{A} : \ d_0 \mid n(N-n)\}$: the condition $d_0 \mid n(N-n)$ splits into the $2^{\omega(d_0)}$ disjoint conditions $d_1 \mid n$, $d_2 \mid N-n$ with $d_1 d_2 = d_0$, of densities $h(d_1)h(d_2)$, and $\sum_{d_1 d_2 = d_0} h(d_1)h(d_2) = \omega_0(d_0)$. For the primes of $P_1 = P/P_0$ we use the Rosser–Iwaniec linear sieve weights [9] $\lambda_d^{\pm}$, $d \mid P_1$, of level $D_A = N^{\theta_A}$ on the side n and $\lambda_e^{\pm}$, $e \mid P_1$, of level $D_B = N^{\theta - \theta_A - \varepsilon}$ on the side $N - n$. Under (23) the main-term sums of these weights are asymptotically equal to their sieve functions, in both directions: with $V_1 = \prod_{p \mid P_1}(1 - h(p)) \asymp \log z_0 / \log z$ and $s = \log D / \log z$,

$$\sum_{d \mid P_1} \lambda_d^{+} h(d) = V_1\,(F_1(s) + o(1)), \qquad \sum_{d \mid P_1} \lambda_d^{-} h(d) = V_1\,(f_1(s) + o(1)), \tag{24}$$

see [2, Section 3] and [9]; the $o(1)$ depends only on the constant in (23), so it is uniform in the sifting set and in particular in the lower cut z_0. Since only the primes below z_0 are removed from the sifting range, $s_A = \log D_A / \log z = \theta_A/\alpha$ and $s_B = (\theta - \theta_A - \varepsilon)/\alpha$ are as in the statement (up to the ε, which is harmless as F_1, f_1 are continuous).

The sieve inequality. For $n \in \mathcal{A}$ put $c^{\pm} = \sum_{d_0 \mid n(N-n)} \rho_{d_0}^{\pm}$, $a^{\pm} = \sum_{d \mid n} \lambda_d^{\pm}$, $b^{\pm} = \sum_{e \mid N-n} \lambda_e^{\pm}$. Then $1_{(n(N-n),P)=1} = 1_{(n(N-n),P_0)=1}\, 1_{(n,P_1)=1}\, 1_{(N-n,P_1)=1}$, and the vector sieve inequality of Brüdern and Fouvry [2] for three factors (obtained from $ab \ge a^- b^+ + a^+ b^- - a^+ b^+$ by applying it twice) gives

$$1_{(n(N-n),P)=1} \ \ge \ c^- a^+ b^+ + c^+ a^- b^+ + c^+ a^+ b^- - 2c^+ a^+ b^+. \tag{25}$$

Multiply by $w(n) \ge 0$ and sum over $n \in \mathcal{A}$. A typical term is

$$\sum_{d_0, d, e} \rho_{d_0}^{-} \lambda_d^{+} \lambda_e^{+} \sum_{\substack{n \in \mathcal{A} \\ d_0 \mid n(N-n),\ d \mid n,\ e \mid N-n}} w(n) = \sum_{d_0, d, e} \rho_{d_0}^{-} \lambda_d^{+} \lambda_e^{+} \sum_{d_1 d_2 = d_0} S(d_1 d, d_2 e),$$

and $S(d_1d, d_2e) = Xh(d_1)h(d_2)h(d)h(e) + R(d_1d, d_2e)$ if $(d, e) = 1$, while $S(d_1d, d_2e) = 0$ if $(d, e) > 1$ (the pairs (d_1, d_2) and (d_0, de) are automatically coprime). Hence the main term of this term is

$$X\Big(\sum_{d_0} \rho^-_{d_0}\omega_0(d_0)\Big)\,\Sigma^{++}, \qquad \Sigma^{\pm\pm'} = \sum_{\substack{d,e\mid P_1\\(d,e)=1}} \lambda_d^{\pm}\lambda_e^{\pm'}h(d)h(e),$$

and similarly for the other three terms of (25).

The coprimality condition. Removing the condition $(d, e) = 1$ from $\Sigma^{\pm\pm'}$ costs

$$\Big|\sum_{\substack{d,e\mid P_1\\(d,e)>1}} \lambda_d\lambda_e h(d)h(e)\Big| \le \sum_{p\mid P_1} h(p)^2\Big(\sum_{d\mid P_1} h(d)\Big)^2 \le \sum_{p\ge z_0}\frac{4}{p^2}\prod_{p\mid P_1}(1+h(p))^2$$
$$\ll \frac{1}{z_0}\Big(\frac{\log z}{\log z_0}\Big)^2 \ll \frac{V_1^2}{z_0}\Big(\frac{\log z}{\log z_0}\Big)^4 = o(V_1^2),$$

because $h(p) < 2/p$, $|\lambda_d| \le 1$, $\prod_{p\mid P_1}(1 + h(p)) \asymp V_1^{-1} \asymp \log z/\log z_0$, and $(\log z/\log z_0)^4/z_0 \ll (\log N)^{-1}$ for $z_0 = (\log N)^5$. Hence, by (24),

$$\Sigma^{-+} = V_1^2\big(f_1(s_A)F_1(s_B) + o(1)\big), \qquad \Sigma^{+-} = V_1^2\big(F_1(s_A)f_1(s_B) + o(1)\big),$$
$$\Sigma^{++} = V_1^2\big(F_1(s_A)F_1(s_B) + o(1)\big).$$

Together with the fundamental lemma for the $\rho^{\pm}$ sums, the main term of the right-hand side of (25) is

$$X\,V_0V_1^2\,\big(F_1F_1 + f_1F_1 + F_1f_1 - 2F_1F_1 + o(1)\big) = X\,V_0V_1^2\,(L + o(1)),$$

with the arguments s_A, s_B understood. Finally

$$V_0V_1^2 = \prod_{\substack{2<p<z_0\\p\nmid N}}(1-2h(p))\prod_{\substack{z_0\le p<z\\p\nmid N}}(1-h(p))^2 = V\prod_{z_0\le p<z}\frac{(1-h(p))^2}{1-2h(p)} = V\Big(1+O\Big(\frac{1}{z_0}\Big)\Big),$$

since each factor of the last product is $1 + O(p^{-2})$.

Remainders. The moduli occurring are the coprime square-free pairs $(D_1, D_2) = (d_1d, d_2e)$ with $D_1D_2 \le D_0D_AD_B \le N^{\theta-\varepsilon/2} \le N^{1/2-\varepsilon/2}$ for large N, and each pair arises from exactly one triple (d_0, d, e) (d_1 and d_2 are the parts of D_1, D_2 composed of primes below z_0). Since $|\rho^{\pm}\lambda^{\pm}\lambda^{\pm}| \le 1$, Theorem 3.2 with $\varepsilon/2$ in place of ε shows that the remainders contribute $O(N^{1-\varepsilon/6}) = o(XV)$. This gives $\sum_{(n(N-n),P)=1} w(n) \ge (L + o(1))XV$.

The weights. For the Richert weights we need an upper bound for $\sum w(n)$ over $n \in \mathcal{A}$ with $(n(N - n), P) = 1$ and $p \mid n$, for a fixed prime $z \le p < y_1$, $p \nmid N$, $p = N^u$. Here $1_{(n(N-n),P)=1} \le c^+a^+b^+$, with Rosser upper-bound weights of total level $N^{\theta-u-\varepsilon}$; the subsequence $\{n \in \mathcal{A} :\ p \mid n\}$ has density $h(p)h(d)h(e)$ on $d \mid n$, $e \mid N - n$ (and $p \nmid de$ automatically, as $p \ge z$), and the same argument bounds its sifted weight by $(U(\theta - u) + o(1))\,h(p)\,XV$, the split of the level between the two sides being free. The remainders are again controlled by Theorem 3.2, since the number of ways of writing a modulus as $p \cdot D_1 \cdot D_2$ is at most $d_3(pD_1D_2)$, which absorbs the sum over p. Since $\sum_{p\equiv 1\,(4),\ N^u\le p<N^{u+\mathrm{d}u}} h(p) = (1 + o(1))\,\mathrm{d}u/u$ by the prime number theorem in progressions ($h(p) = 2/p + O(p^{-2})$ on half of the primes), and the side $N - n$ contributes the same amount, the weighted sum is bounded by $(R + o(1))XV$, the factor 2 in R accounting for the two sides.

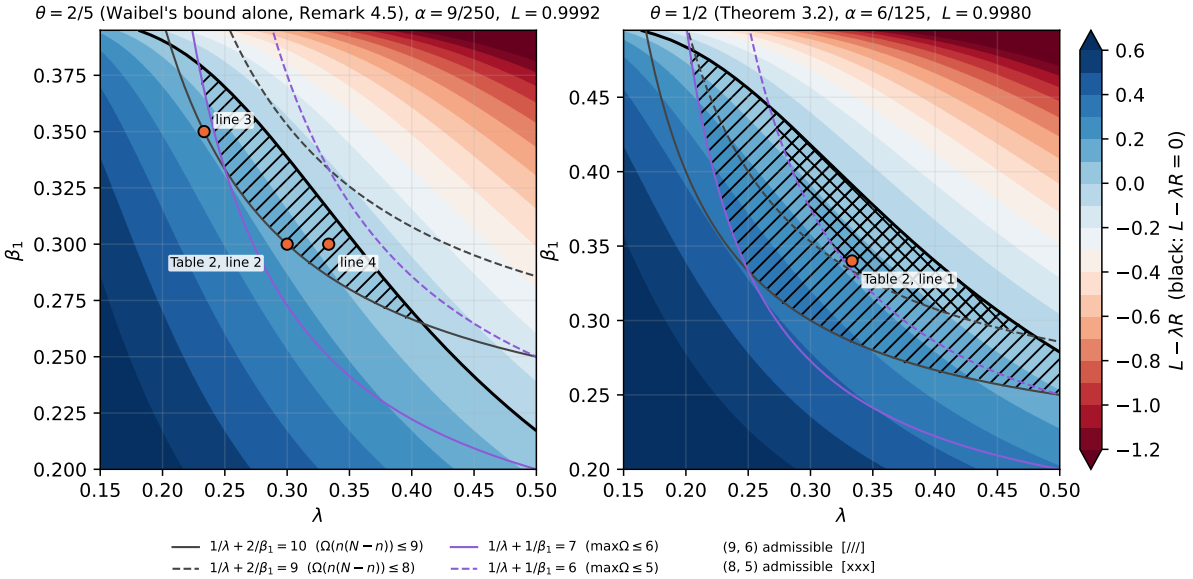


FIGURE 1. $L - \lambda R$ as a function of (λ, β_1) for $\theta = 2/5$, $\alpha = 9/250$ (left) and $\theta = 1/2$, $\alpha = 6/125$ (right), with the best split θ_A in each case. Black: $L - \lambda R = 0$. Grey and violet curves: $1/\lambda + 2/\beta_1 = 10$, 9 and $1/\lambda + 1/\beta_1 = 7$, 6; the bound $\Omega(n(N-n)) \le k$ (resp. $\max \Omega \le k'$) holds above the corresponding curve. Hatched: parameters with $L - \lambda R > 0$ giving $(9, 6)$ (single hatching) and $(8, 5)$ (cross hatching). Dots: the parameter sets of Table 2. Floating point (`plot_sieve_landscape.py`), for orientation only.

The constant. By (17), Lemma 3.1 and (19),

$$XV = 8N \prod_{p \mid N^b} \frac{\beta_p^*}{1 - p^{-2}} \prod_{\substack{2<p<z \\ p \nmid N}} (1 - 2h(p)) = 8N\, \mathfrak{S}_r(N) \prod_{2<p<z} (1 - 2h(p)) \prod_{\substack{p \mid N \\ p \ge z}} (1 - 2h(p)),$$

and the last product is $1 + O(\omega(N)/z)$. Writing $1 - 2h(p) = (1 - h(p))^2 \cdot \frac{1-2h(p)}{(1-h(p))^2}$, where the second factor is $1 - O(p^{-2})$, Mertens' theorem in arithmetic progressions gives $\prod_{2<p<z}(1-h(p)) \sim c_h / \log z$ and hence $\prod_{2<p<z}(1-2h(p)) \sim c_{2h}/(\log z)^2$ with $c_{2h} = \kappa_0 c_h^2$, $\kappa_0 = \prod_{p>2}(1-2h(p))/(1-h(p))^2$; the evaluation $c_{2h} \in [1.66485, 1.66487]$ is described in Appendix A.3. □

4.1. **Numerical verification.** The parameters α, λ, β_1 are chosen so that $L - \lambda R > 0$ with $1/\lambda + 2/\beta_1 \le 9$ and $1/\lambda + 1/\beta_1 \le 6$, which by Lemma 4.2 gives the bounds 8 and 5. Figure 1 shows the function $L - \lambda R$ in the (λ, β_1) plane for the levels 2/5 and 1/2 and the α of Table 2, together with the curves $1/\lambda + 2/\beta_1 = k + 1$ and $1/\lambda + 1/\beta_1 = k' + 1$: at the level 1/2 the region giving $(8, 5)$ meets the region $L - \lambda R > 0$; at the level 2/5 only the region giving $(9, 6)$ does. The figure is drawn with floating-point sieve functions and is not part of the proof; the four parameter sets of Table 2 are certified in interval arithmetic (Lemma 4.4, Appendix A).

Lemma 4.4. *The bounds of Table 2 hold.*

Proof. The script `certify_modular.py` carries out a finite computation with directed rounding; the scheme is described in full in Appendix A. The rational numbers $1/\lambda + 2/\beta_1$ and $1/\lambda + 1/\beta_1$ are computed exactly. □

θ	α	λ	β_1	$1/\lambda+2/\beta_1$	$1/\lambda+1/\beta_1$	$L\geq$	$R\leq$	$L-\lambda R\geq$
1/2	6/125	1/3	17/50	151/17	101/17	0.997718	2.498893	0.164754
2/5	9/250	3/10	3/10	10	20/3	0.998974	2.816690	0.153967
2/5	4/125	7/30	7/20	10	50/7	0.999637	3.471422	0.189639
2/5	9/250	1/3	3/10	29/3	19/3	0.998974	2.816690	0.060078

TABLE 2. Certified bounds. The split θ_A is chosen on a grid near $\theta/2$ ($\theta_A=0.2497$ in the first line, 0.1998 for $\alpha=9/250$, 0.1992 for $\alpha=4/125$); any $\theta_A\in(0,\theta)$ is admissible in Proposition 4.3. The first line, at the level 1/2 of Theorem 3.2, gives $\Omega(n(N-n))\leq 8$ and $\max(\Omega(n),\Omega(N-n))\leq 5$ (Theorem 1.1). The other three lines are at the level 2/5 that Lemma 2.1 alone provides (Remark 4.5): the second gives $\Omega(n(N-n))\leq 9$ and $\max(\Omega(n),\Omega(N-n))\leq 6$; the third $\Omega(n(N-n))\leq 9$ with a larger constant; the fourth the same bounds as the second with $\lambda=1/3$.

Proof of Theorem 1.1. Take the first line of Table 2: $\theta=1/2$, $\alpha=6/125$, $\lambda=1/3$, $\beta_1=17/50$, so that $1/\lambda+2/\beta_1=151/17<9$ and $1/\lambda+1/\beta_1=101/17<6$. By Proposition 4.3 and Lemma 4.4, $W\geq(0.164754+o(1))XV$. Since $w_n\leq 1$ and $w(n)\geq 0$, the sum of $w(n)$ over the n with $(n(N-n),P)=1$ and $w_n>0$ is at least W; removing the exceptional n of Lemma 4.1 costs $O(N^{1-\alpha+\varepsilon})$. By Lemmas 4.1 and 4.2 every remaining n has n, $N-n$ sums of two squares, $\Omega(n(N-n))\leq 8$, $\max(\Omega(n),\Omega(N-n))\leq 5$ and $w(n)=16\cdot 2^{\Omega(n(N-n))}\leq 2^{12}$. Hence the number of such n is at least

$$\big(0.164754+o(1)\big)\,2^{-12}\cdot 8\,\mathfrak{S}_r(N)\,c_{2h}\,\frac{N}{\alpha^2(\log N)^2}\;\geq\;0.23\,\mathfrak{S}_r(N)\frac{N}{(\log N)^2}$$

for large N, since $0.164754\cdot 2^{-9}\cdot c_{2h}\cdot(125/6)^2\geq 0.232$. Finally $\mathfrak{S}_r(N)\geq 1$, every factor $(p-1)/(p-2)$ exceeding 1. □

Remark 4.5 (Waibel's bound alone). Without Theorem 1.2, i.e. with Lemma 2.1 as the only information on the norm, Theorem 3.2 gives the level 2/5, and the second line of Table 2 ($\theta=2/5$, $\alpha=9/250$, $\lambda=\beta_1=3/10$, $L-\lambda R\geq 0.153967$) gives, by the same proof with $w(n)\leq 2^{13}$, the bounds $\Omega(n(N-n))\leq 9$ and $\max(\Omega(n),\Omega(N-n))\leq 6$ with the constant 0.19 (since $0.153967\cdot 2^{-10}\cdot c_{2h}\cdot(250/9)^2\geq 0.193$) and primes exceeding $N^{9/250}$. This was Theorem 1.1 of the first two versions of this paper. Figure 1 shows that at the level 2/5 the pair $(8,5)$ is out of reach of these weights, so that the improvement from $(9,6)$ to $(8,5)$ is entirely due to Theorem 1.2. In the other direction, Part II [4, Section 7] replaces the vector sieve used here by the two-dimensional sieve of Diamond, Halberstam and Richert, for which $n(N-n)$ is the natural object (sieving limit $\beta_2=4.2665$ against $2+2\sqrt{e}=5.297$ for the vector sieve), and obtains $(7,5)$ from the same level 1/2, with the sieve functions F_2, f_2 enclosed in interval arithmetic as in Appendix A.1.

5. Numerical evidence

Nothing in this section is used in the proofs of Sections 2–4. Its purpose is to measure how the error term of Theorem 2.3 actually depends on mm'. Historically, these data were computed before Theorem 1.2 was available, and the absence of growth in mm' that they show was formulated as a conjecture in the first two versions of this paper; Theorem 1.2 proves it, and Part II [4] measures the norm $\|G_{m,m'}\|^2$ directly from the same data.

5.1. What was computed.

The quantities $T(m,m';N)$ for all $N\leq X$ are obtained at once by a fast Fourier transform convolution of the sequences $a\mapsto r(a/m)$ and $b\mapsto r(b/m')$ (with $r(x)=0$ for $x\notin\mathbb{Z}$), and $M(m,m';N)$ from (5)–(7), with the 2-adic density $\beta_2(N)$ obtained from

exact counts of solutions of $Q(x) \equiv N$ modulo 2^ν, $\nu = v_2(N) + 6$, and checked to be unchanged when ν is replaced by $\nu+1$ (script `sup_error_scan.py`). We record the statistics of $(T-M)/\sqrt{N}$ over all $N \in (X/2, X]$ with

$$(N, mm') = 1 \quad \text{and} \quad v_2(N) \le 3. \tag{26}$$

The condition $(N, mm') = 1$ is the hypothesis of Theorem 2.3; it isolates the cusp form $G_{m,m'}$ from the reduction of Lemma 2.5, so that what is measured is directly the quantity $a_G(N)$ bounded in Lemma 2.2, and for $m \equiv m' \pmod 4$ it makes the main term the explicit product (9) (the sample contains pairs with $m \not\equiv m' \pmod 4$ as well, e.g. $(3,5)$ and $(61,67)$; since β_2 is obtained from exact counts, no hypothesis modulo 4 is needed, and the value $1/2$ of Section 2 is reproduced for these pairs at $N \equiv 2 \pmod 4$). The condition $v_2(N) \le 3$ keeps $\nu \le 9$, so that β_2 is a function of N modulo 2^9 that can be tabulated exactly. It includes the case $N \equiv 2 \pmod 4$ used in the sieve and the case of odd N; the values $v_2(N) \in \{2,3\}$ lie outside the range of Theorem 2.3 and are included as additional data only.

Sample. For $X = 2^{21}$ we computed 21 pairs (m, m') with $1 \le mm' \le 1022117$, each over all admissible N, which is between $5 \cdot 10^5$ and 10^6 values of N per pair. For $X = 2^{23}$ we computed 9 of these pairs, with between $2 \cdot 10^6$ and $4 \cdot 10^6$ values of N per pair. Table 3 shows a representative selection; the complete outputs are in the repository.

Precision. The convolutions are computed in IEEE double precision and rounded to the nearest integer; the round-off of the FFT at these sizes is far below $1/2$, and the identity $T = M$ for $mm' \in \{1,3\}$ below checks the whole pipeline end to end. Unlike the sieve computation of Section 4, this computation is *not* carried out in interval arithmetic, and it is not meant to be: its output is statistical evidence, not a step in a proof.

5.2. **Consistency checks.** Three checks, each with an arithmetic reason, are built into the computation.

(1) For $mm' \in \{1,3\}$ the level is 4 or 12, and $S_2(\Gamma_0(4)) = S_2(\Gamma_0(12)) = 0$. Hence $\Theta = E$, i.e. $T(m,m';N) = M(m,m';N)$ for every N. The computed errors are identically 0, which checks the FFT, the formula (5) and the 2-adic density at once; for $(m,m') = (1,3)$ this includes the value $\beta_2 = 1/2$ at $N \equiv 2 \pmod 4$.

(2) For $(m,m') = (1,5)$ and $(9,1)$ the levels are 20 and 36, and $S_2(\Gamma_0(20))$, $S_2(\Gamma_0(36))$ are each spanned by one newform f, with $a_f(2) = 0$ (in both cases 4 divides the level and the character is trivial). Thus $G = c\,f$ for a constant c, and for $N \equiv 2 \pmod 4$ multiplicativity gives $a_G(N) = c\,a_f(2)\,a_f(N/2) = 0$, so $T = M$ for all such N, while for odd N the error is non-zero. Both features are observed.

(3) The density (7) was compared with exact counts of solutions modulo p^2 for all primes $p \le 13$ dividing mm' and all N prime to p; they agree.

5.3. **Observations.** Two features stand out. First, the root mean square decreases like $(\log mm')(mm')^{-1/2}$: over the whole sample the quotient $\mathrm{rms} \cdot (mm')^{1/2}/\log mm'$ lies between 2.2 and 3.1 (Figure 2). This is what Theorem 1.2 predicts. By the Rankin–Selberg method, the mean square of $a_G(n)/\sqrt{n}$ over $n \le Y$ is asymptotic to an absolute constant times $\|G\|^2/\mathrm{vol}(\Gamma_0(4mm')\backslash\mathbb{H})$, the volume is $\asymp mm'$, and $\|G_{m,m'}\|^2$ grows only like $(\log mm')^2$ (Theorem 1.2; numerically $\approx 0.37(\log mm')^2$, [4, Section 5]). In the first two versions of this paper the decay of the root mean square was read off as $(mm')^{-1/4}$ and interpreted as Lemma 2.1 being sharp; that reading was wrong, as both the fit and Theorem 1.2 show. Part II contains the precise measurement, including an exact check at the level 20, where $S_2(\Gamma_0(20))$ is one-dimensional, $G_{1,5} = \frac{8}{3} f_{20}$ with $f_{20} = \eta(2z)^2\eta(10z)^2$, and $\|f_{20}\|^2 = 0.08135$ both from the period lattice of the

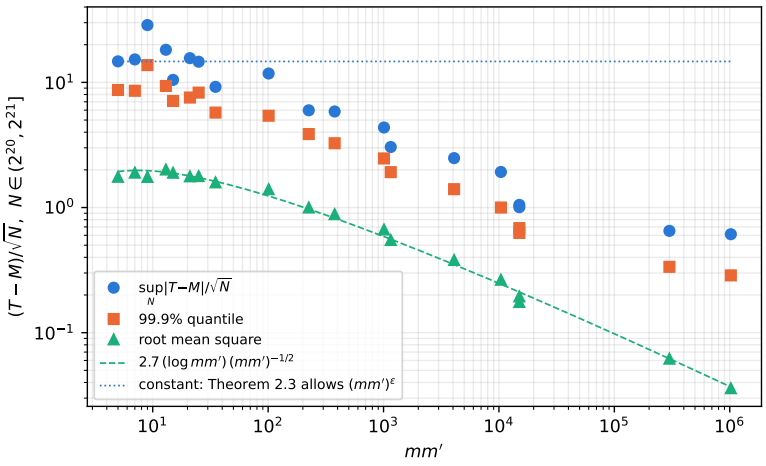


FIGURE 2. The data of Table 3 for $X = 2^{21}$ (all pairs computed), with the curve $2.7\,(\log mm')(mm')^{-1/2}$ (the median of $\mathrm{rms}\cdot(mm')^{1/2}/\log mm'$ over the sample) and a constant for comparison. The supremum does not grow with the level; the root mean square decreases like $(\log mm')(mm')^{-1/2}$.

(m,m')	mm'	rms	99.9% quantile	sup, $X=2^{21}$	sup, $X=2^{23}$
$(1,5)$	5	1.76	8.7	14.7	17.3
$(9,1)$	9	1.76	13.7	28.7	33.8
$(3,5)$	15	1.90	7.1	10.5	12.2
$(1,101)$	101	1.40	5.4	11.8	12.1
$(13,29)$	377	0.89	3.3	5.9	6.8
$(1,1009)$	1009	0.67	2.5	4.4	5.3
$(61,67)$	4087	0.38	1.4	2.5	2.4
$(101,103)$	10403	0.27	1.0	1.9	1.8
$(1,15015)$	15015	0.18	0.6	1.0	–
$(1009,1013)$	1022117	0.036	0.29	0.61	0.55

TABLE 3. Statistics of $(T(m,m';N)-M(m,m';N))/\sqrt{N}$ over all $N\in(X/2,X]$ satisfying (26) ($\approx 10^6$ values for $X=2^{21}$, $4\cdot10^6$ for $X=2^{23}$; the run with $X=2^{23}$ covers a selection of the pairs).

elliptic curve 20a1 and from the Rankin–Selberg asymptotics of the data. (Waibel's lower bound [18, Theorem 3] gives $\|G\|^2 \gg 1/\min(m,m')$ for our forms.)

Second, the supremum over N does not grow with mm' at all—it decreases—while Theorem 2.3 allows a factor $(mm')^\varepsilon$, and it is larger than the root mean square by a factor of about $(mm')^{1/2}$ at the top of the range. The gap between the two statistics is the Cauchy–Schwarz step in Lemma 2.2. Expanding in an orthogonal basis of Hecke eigenforms (Section 5.5), $a_G(N)$ is a sum of $\asymp mm'$ terms $\langle G,f\rangle a_f(N)/\|f\|^2$; Cauchy–Schwarz bounds this sum by $\|G\|$ times the square root of the Petersson sum $\sum_f |a_f(N)|^2/\|f\|^2 \asymp N$, and it is attained only if the phases of the individual terms align with those of $\overline{a_f(N)}$. The data are consistent with the opposite: the terms behave like a sum with essentially random signs, whose size is about $(mm')^{-1/2}$ times the Cauchy–Schwarz bound in the root mean square (the square root of the number of terms), and does not exceed the size for fixed level in the supremum. Figure 3 shows the distribution of $(T-M)/\sqrt{N}$, divided by its root mean square, for four levels between 404 and 16348: the four

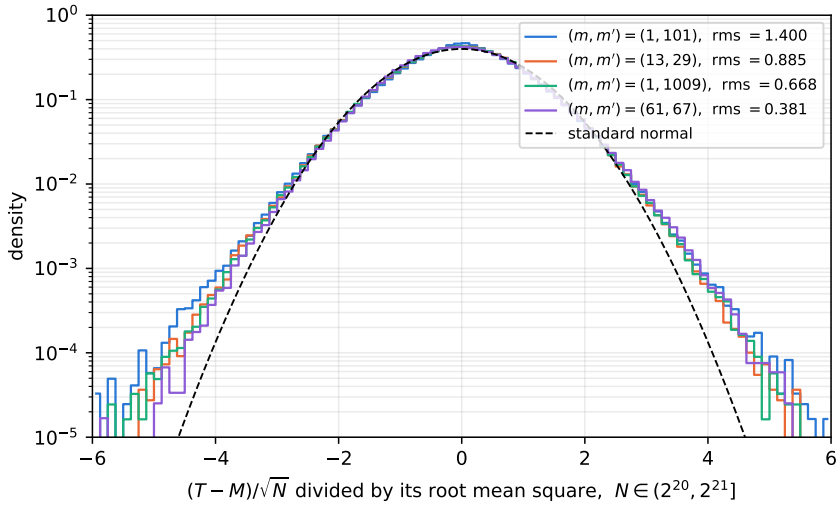


FIGURE 3. Empirical density of $(T(m,m';N)-M(m,m';N))/\sqrt{N}$ divided by its root mean square, over all $N\in(2^{20},2^{21}]$ satisfying (26) (about 10^6 values per pair), for four pairs (m,m'), on a logarithmic scale, against the standard normal density (dashed). Script `plot_error_hist.py`; floating point, not certified. For mm' of size 10^6 the main term M is of size 30 while T is a multiple of 8, and the distribution is dominated by this discreteness; such pairs are not shown.

empirical densities coincide, are symmetric and Gaussian in the bulk, with tails somewhat heavier than Gaussian (the kurtosis is 3.5–3.7 and about 0.55% of the values exceed three standard deviations, against 0.27% for the normal law), as one expects from the divisor-type factor in Deligne's bound. The growth in N between $X=2^{21}$ and 2^{23} is by a factor 1.0–1.2, as expected from Deligne's bound (the suprema are attained at N with 16–96 divisors).

5.4. **Status of the numerical evidence.** We stress the logical status of these data. They consist of finitely many pairs (m,m') and finitely many $N\le 2^{23}$, and no finite computation can establish a bound that is uniform in m,m' and N; in particular the computations do not bear on Theorem 1.1, Theorem 3.2 or Theorem 2.3, which are proved independently of them. The data are compatible with Theorem 2.3, and they suggest more: that $|T-M|\ll N^{1/2+\varepsilon}(mm')^{-1/2+\varepsilon}$ holds for *every* N with $(N,mm')=1$, not only in the mean square over N, where Part II proves it ([4, Theorem 1.2], which gives the level of distribution $1-\varepsilon$ for almost all N). Such a uniform bound would give the level of distribution $1-\varepsilon$ for every N, and with it the bounds 5 and 4 in Theorem 1.1 ([4, Section 4]). For almost all N this is not where the interest lies, since the circle method gives much more there: by the method of Montgomery and Vaughan [14], almost all $N\equiv 2 \pmod 4$ are sums of two primes $\equiv 1 \pmod 4$. It is for every N, where the circle method gives nothing, that the sieve has something to say, and there the level is $1/2$ and the bounds are 8 and 5. The data do not prove the uniform bound, and they give no information about N or mm' beyond the range computed.

5.5. **An automorphic interpretation.** The following discussion is heuristic and is not used anywhere in the paper. Let $L=4mm'$ and let $\mathcal{B}^*$ be an orthogonal basis of $S_2(\Gamma_0(L))$ consisting of eigenforms of all T_n, $(n,L)=1$; by the proof of Lemma 2.2 it may be taken inside the spaces $S(f)$, so every element $\varphi\in\mathcal{B}^*$ has $a_\varphi(n)=\lambda_f(n)\,n^{1/2}a_\varphi(1)$ for $(n,L)=1$, where f is the underlying newform and $\lambda_f(n)=a_f(n)n^{-1/2}$. Since E is orthogonal to cusp forms, $\langle\Theta,\varphi\rangle=\langle G,\varphi\rangle$, and for

$(N, L) = 1$

(27)

$$T(m, m'; N) - M(m, m'; N) = a_G(N) = \sum_{\varphi \in \mathcal{B}^*} \frac{\langle \Theta, \varphi \rangle}{\|\varphi\|^2} a_\varphi(N) = N^{1/2} \sum_{\varphi \in \mathcal{B}^*} \frac{\langle \Theta, \varphi \rangle \, a_\varphi(1)}{\|\varphi\|^2} \lambda_f(N).$$

The coefficients $\langle \Theta, \varphi \rangle$. The form θ^2 is the weight-one Eisenstein series of level 4 and character χ, with L-function $\sum r(n) n^{-s} = 4\zeta(s)L(s, \chi)$. Writing $\theta^2(m'z)$ as a combination of weight-one Eisenstein series for the cusps of $\Gamma_0(4m')$ and unfolding (the Rankin–Selberg method in Shimura's form [15], with Hecke's trick at the holomorphic point), $\langle \theta^2(m\,\cdot)\theta^2(m'\,\cdot), \varphi \rangle$ becomes a combination of values at the centre of Dirichlet series of the shape $\sum_n \overline{\lambda_f(n)}\, r(n)\, n^{-s}$, twisted by the local data at $2mm'$. Since $r/4 = 1 * \chi$ is the convolution of two characters, $\sum_n \lambda_f(n) r(n) n^{-s} = 4L(s, f)L(s, f \otimes \chi)/L(2s, \chi)$ up to Euler factors at the primes dividing L, and one expects

$$\langle \Theta, \varphi \rangle = \mathcal{E}_\varphi(m, m') \; L(\tfrac{1}{2}, f)\, L(\tfrac{1}{2}, f \otimes \chi), \tag{28}$$

with a factor $\mathcal{E}_\varphi(m, m')$ built from local factors at the primes dividing $2mm'$ and from the normalisation of φ. We have not worked out $\mathcal{E}_\varphi$ here, and (28) should be read as a statement about the structure of the coefficients, not as an identity we use; Part II [4, Theorem 6.7] proves it for newforms φ of level $4mm'$ with mm' square-free, with $\mathcal{E}_\varphi(m, m') = \frac{4}{\pi^2} \chi(m') \eta_{m'}(\varphi)$, $\eta_{m'}$ the Atkin–Lehner eigenvalue, and $L(\frac{1}{2}, \cdot)$ written as $L(1, \cdot)$ in the classical normalisation. (In particular, it is the product of the two central values that enters linearly, as in the Rankin–Selberg method, not a square root of it as in Waldspurger-type formulas for half-integral weight.)

Heuristic reformulation. Assuming the structural factorisation (28), which we have not proved, and inserting it into (27), the cuspidal error becomes

$$\begin{aligned} T(m, m'; N) - M(m, m'; N) &= N^{1/2}\, \mathcal{S}(m, m'; N), \\ \mathcal{S}(m, m'; N) &= \sum_{\varphi \in \mathcal{B}^*} \omega_\varphi \, L(\tfrac{1}{2}, f)\, L(\tfrac{1}{2}, f \otimes \chi)\, \lambda_f(N), \end{aligned} \tag{29}$$

with weights $\omega_\varphi = \mathcal{E}_\varphi(m, m')\, a_\varphi(1)/\|\varphi\|^2$, i.e. $N^{1/2}$ times a twisted first moment over a family of $\asymp mm'$ forms of weight 2 and level $4mm'$. In these terms, Theorem 2.3 is what one gets by bounding $\mathcal{S}$ through Cauchy–Schwarz, with the second moment of the coefficients $\langle \Theta, \varphi \rangle$ controlled by $\|G\|^2 \ll (mm')^\varepsilon$ (Theorem 1.2) and the second moment of $\lambda_f(N)$ controlled by the Petersson formula; this gives $\mathcal{S} \ll N^\varepsilon (mm')^\varepsilon$ and loses all cancellation among the terms of (29). The data of Section 5 say that in the root mean square over N the sum $\mathcal{S}$ is smaller than this by the factor $(mm')^{-1/2}$, the square root of the number of terms, which is what random signs would give; Part II proves this mean-square statement [4, Theorem 1.2] by a covering argument that does not use the spectral expansion. The data say moreover that no individual N in the range computed exceeds the root mean square by more than a bounded factor. A proof of $\mathcal{S}(m, m'; N) \ll N^\varepsilon (mm')^{-1/2+\varepsilon}$ for every N would have to exploit the oscillation of $\lambda_f(N)$ across the family; since the summands are not of constant sign (because of $\lambda_f(N)$ and of the local factors), it cannot come from positivity.

A possible approach. The standard route to first moments of this kind is to write $L(\frac{1}{2}, f)L(\frac{1}{2}, f \otimes \chi)$ by an approximate functional equation of length about the square root of the analytic conductor $\asymp (mm')^2$, i.e. about mm', to combine $\lambda_f(a)\lambda_f(N)$ by the Hecke relations, and to average over the family with the Petersson formula (as in the work of Iwaniec and Sarnak [10] on first moments with harmonic weights). The diagonal terms are then explicit, and the off-diagonal terms are sums of Kloosterman sums $S(aN, b; c)$ with $c \equiv 0 \pmod{4mm'}$, to which Weil's bound applies. The main obstacle is that in the range relevant for a level of distribution beyond $1/2$ one has

$mm' \le N^{1-\varepsilon}$, so the twist $\lambda_f(N)$ is long compared with the level, and the off-diagonal terms are not small without further cancellation in the sums over a, b and c (for instance from the Kuznetsov formula and bounds for sums of Kloosterman sums). We have not attempted this.

Appendix A. The numerical verification of Lemma 4.4 and of the constants

This appendix describes precisely what is computed by the two scripts `certify_modular.py` and `singular_constant.py`. Let $h_0 = 10^{-3}$ and $s_i = ih_0$. For the closed forms, all interval computations use the interval type `mpmath.iv` with 80-bit precision; for the recurrences they use IEEE double precision with one outward `nextafter` step after every arithmetic operation; we write $\lfloor x \rfloor_\downarrow$ and $\lceil x \rceil^\uparrow$ for the outward-rounded results.

A.1. **The linear sieve functions.** The functions F_1, f_1 are determined by

$$F_1(s) = \frac{2e^\gamma}{s} \quad (0 < s \le 3), \qquad f_1(s) = \frac{2e^\gamma \log(s-1)}{s} \quad (2 \le s \le 4),$$
$$(sF_1(s))' = f_1(s-1) \quad (s > 3), \qquad (sf_1(s))' = F_1(s-1) \quad (s > 2),$$

and F_1 is decreasing, f_1 increasing on their domains [6, Chapter 11]. On the grid points $s_i \le 3$ (resp. $2 < s_i \le 4$) enclosures $[F_i^-, F_i^+]$ (resp. $[f_i^-, f_i^+]$) are obtained by evaluating the closed forms in interval arithmetic. For larger s we integrate the delay equations over one grid cell. With $G_F(s) = sF_1(s)$, $G_f(s) = sf_1(s)$ and $j = i - 1000$ (so $s_j = s_i - 1$),

$$G_F(s_{i+1}) - G_F(s_i) = \int_{s_i}^{s_{i+1}} f_1(s-1)\,\mathrm{d}s \in \big[h_0\, f_1(s_j),\ h_0\, f_1(s_{j+1})\big],$$

because f_1 is increasing, and

$$G_f(s_{i+1}) - G_f(s_i) = \int_{s_i}^{s_{i+1}} F_1(s-1)\,\mathrm{d}s \in \big[h_0\, F_1(s_{j+1}),\ h_0\, F_1(s_j)\big],$$

because F_1 is decreasing. Replacing the values on the right by their enclosures gives the one-step recursion

$$G_F^-(s_{i+1}) = \lfloor G_F^-(s_i) + h_0^- f_j^- \rfloor_\downarrow, \qquad G_F^+(s_{i+1}) = \lceil G_F^+(s_i) + h_0^+ f_{j+1}^+ \rceil^\uparrow,$$
$$G_f^-(s_{i+1}) = \lfloor G_f^-(s_i) + h_0^- F_{j+1}^- \rfloor_\downarrow, \qquad G_f^+(s_{i+1}) = \lceil G_f^+(s_i) + h_0^+ F_j^+ \rceil^\uparrow,$$

starting from the interval values at $s = 3$ and $s = 4$, and

$$F_{i+1}^- = \big\lfloor G_F^-(s_{i+1})/s_{i+1}^+ \big\rfloor_\downarrow, \qquad F_{i+1}^+ = \big\lceil G_F^+(s_{i+1})/s_{i+1}^- \big\rceil^\uparrow,$$

and similarly for $f_{i+1}^\pm$. Here $[s_i^-, s_i^+]$ is the enclosure of $s_i = i/1000$, and $h_0^- = \lfloor s_{i+1}^- - s_i^+ \rfloor_\downarrow$, $h_0^+ = \lceil s_{i+1}^+ - s_i^- \rceil^\uparrow$ are the lower and upper enclosures of the step $s_{i+1} - s_i$; the lower enclosure is used in the formulas for G^- and the upper one in those for G^+ (all factors $f_j^\pm$, $F_j^\pm$ being non-negative). This is a one-step enclosure, not an iteration to a fixed point: since the delay is exactly 1000 grid steps, the values needed on the right have always been computed earlier. By induction $F_1(s_i) \in [F_i^-, F_i^+]$ and $f_1(s_i) \in [f_i^-, f_i^+]$ for $s_i \le 60$. The enclosures are narrow enough: on the closed-form ranges their width is that of a double-precision number, and beyond $s = 3$, resp. $s = 4$, the one-step monotone bounds widen them to about $2 \cdot 10^{-4}$ within one unit of s (the first-order discretisation error of the Riemann bounds, $O(h_0)$), after which the width stays at that size (Figure 4). The computation gives $F_1(3), f_1(4)$ to six decimals, and $F_1(60) \le 1.000066$, $f_1(60) \ge 0.999935$, in agreement with $F_1, f_1 \to 1$.

Off the grid, monotonicity gives

$$F_1(s) \le F^+_{\lfloor s/h_0 \rfloor - 1}, \qquad F_1(s) \ge F^-_{\lceil s/h_0 \rceil + 1}, \qquad f_1(s) \ge f^-_{\lfloor s/h_0 \rfloor - 1}$$

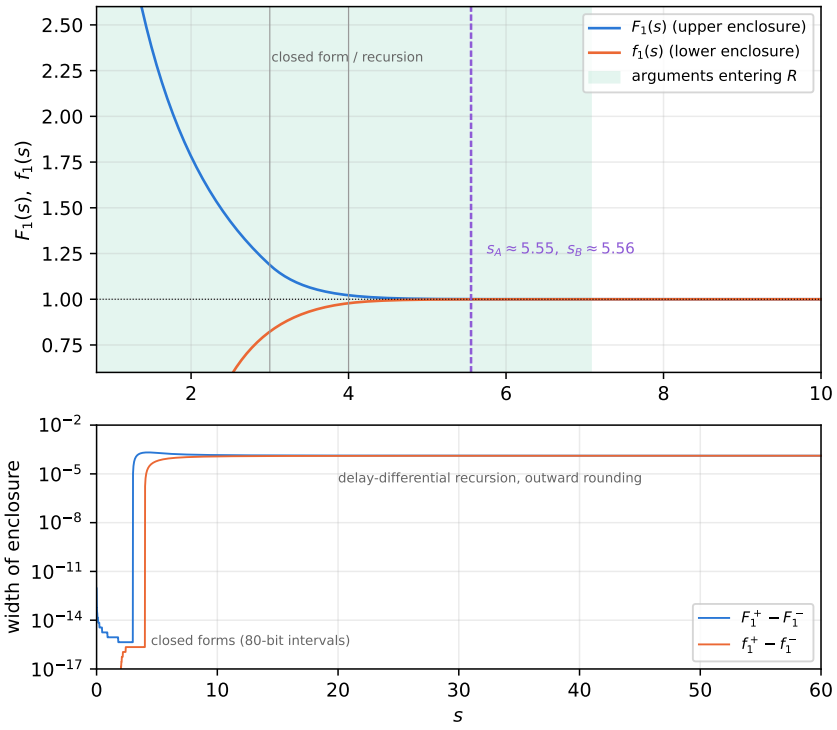


FIGURE 4. Top: the enclosures of F_1 (upper) and f_1 (lower) computed by `certify_modular.py`, the arguments $s_A \approx 5.55$, $s_B \approx 5.56$ at which L is evaluated for the second line of Table 2 ($\theta = 2/5$, $\alpha = 9/250$), and the range of arguments t'/α, $(\theta - u - t')/\alpha$ entering the bound for R (shaded). Bottom: the widths $F_1^+ - F_1^-$ and $f_1^+ - f_1^-$ of the enclosures on the grid, on a logarithmic scale; the jump at $s = 3$, resp. $s = 4$, is the passage from the closed forms to the recursion.

(the extra index shift guards against the rounding of s/h_0), and $F_1(s) \le \lceil 2e^\gamma/s \rceil^\uparrow$ for $s \le 3$.

A.2. **The bounds for L and R.** *The lower bound for L.* We write $L = f_1(s_A)F_1(s_B) + F_1(s_A)\big(f_1(s_B) - F_1(s_B)\big)$. Since $f_1(s_B) - F_1(s_B) \le 0$, a lower bound is

$$L \ge f_1^-(s_A)\, F_1^-(s_B) + F_1^+(s_A)\, \big(f_1^-(s_B) - F_1^+(s_B)\big),$$

with the off-grid bounds of Section A.1 and outward rounding. The split θ_A is a free parameter; the script scans 801 values in $[0.3\theta, 0.7\theta]$ in floating point, keeps the best, and then evaluates the lower bound rigorously at that value. This gives $L \ge 0.997718$ for $\theta = 1/2$, $\alpha = 6/125$ ($\theta_A = 0.2497$, $s_A = 5.2031$, $s_B = 5.2135$) and $L \ge 0.998974$ for $\theta = 2/5$, $\alpha = 9/250$ ($\theta_A = 0.1998$, $s_A = 5.5500$, $s_B = 5.5611$).

The upper bound for R. Divide $[\alpha, \beta_1]$ into $K = 4000$ cells $[u_k, u_{k+1}]$, $u_k = \alpha + k(\beta_1 - \alpha)/K$ (enclosed outward). On a cell:

- the factor $(1 - u/\beta_1)/u$ is decreasing in u, hence at most its value at u_k;
- $U(t)$ is non-increasing in t, because F_1 is decreasing; so $U(\theta - u) \le U(\theta - u_{k+1})$ on the cell;
- $U(t) \le F_1(t'/\alpha)F_1((t - t')/\alpha)$ for every admissible t', since U is a minimum; the script takes the smallest of these upper bounds over 81 values $t' \in [0.3t, 0.7t]$, each evaluated with F_1^+.

Hence

$$R \le 2\sum_{k=0}^{K-1} \Big\lceil \frac{1-u_k/\beta_1}{u_k} \Big\rceil^{\uparrow} \cdot \min_{t'} F_1^+\Big(\frac{t'}{\alpha}\Big) F_1^+\Big(\frac{\theta-u_{k+1}-t'}{\alpha}\Big)\cdot(u_{k+1}-u_k),$$

every operation rounded upwards. For the first line of Table 2 this gives $R \le 2.498893$, and then $L-\lambda R \ge 0.164754$ with λ rounded upwards; for the second line, $R \le 2.816690$ and $L-\lambda R \ge 0.153967$. The other lines of Table 2 are obtained in the same way.

The verification certifies the numerical inequality $L-\lambda R>0$ only. The sieve lemmas behind Proposition 4.3 and the $o(1)$ terms are proved, not computed.

A.3. **The constants c_h and c_{2h}.** Let $P_1(z)=\prod_{p<z,\ p\equiv 1\,(4)}(1-1/p)$. Mertens' theorem $\prod_{p<z}(1-1/p)\sim e^{-\gamma}/\log z$ and the convergence of $\prod_p(1-\chi(p)/p)$ to $L(1,\chi)^{-1}=4/\pi$ give

$$P_1(z)^2=\frac{\prod_{p<z}(1-1/p)\,\prod_{p<z}(1-\chi(p)/p)}{(1-\frac12)\prod_{q<z}(1-q^{-2})}\sim\frac{2e^{-\gamma}}{\kappa\log z},\qquad \kappa=\frac{\pi}{4}\prod_{q\equiv 3\,(4)}\Big(1-\frac{1}{q^2}\Big).$$

Writing $\prod_{2<p<z}(1-h(p))=P_1(z)^2\,C(z)$ with

$$C(z)=\prod_{\substack{p<z\\ p\equiv 1\,(4)}}\Big(1+\frac{2p-1}{(p+1)(p-1)^2}\Big)\prod_{\substack{q<z\\ q\equiv 3\,(4)}}\Big(1-\frac{1}{q(q-1)}\Big)$$

(for split p, $(1-h(p))(1-1/p)^{-2}=p(p^2-p+1)/((p+1)(p-1)^2)$), we get the exact identity $c_h=(2e^{-\gamma}/\kappa)\,C(\infty)$; both infinite products converge absolutely, their factors being $1+O(p^{-2})$.

We compute the products over the primes $p\le P=10^7$ in double precision by summing logarithms, and bound the tails as follows. For $p>P$ the logarithms of the factors are at most $2.01/p^2$ in absolute value (for the split factors, $(2p-1)/((p+1)(p-1)^2)<2.01/p^2$ for $p>10^7$, and for the others $|\log(1-x)|\le x(1+x)$ with $x\le 1.01/p^2$), and

$$\sum_{p>P}\frac{1}{p^2}<\sum_{n>P}\frac{1}{n^2}<\frac{1}{P}.$$

Hence the tails of $\log\kappa$ and of $\log C(\infty)$ lie in $[-1.01/P,0]$ (its factors are $1-q^{-2}$) and $[-2.01/P,2.01/P]$ respectively; the script uses the wider margins $2/P$ for $\log\kappa$ and $4/P$ for $\log C(\infty)$. The rounding error of the double-precision summation over the $\approx 6.6\cdot 10^5$ primes is of relative size 10^{-10}, far inside these margins. This gives

$$\kappa\in[0.67238629,\,0.67238642],\qquad C(\infty)\in[0.89899105,\,0.89899177],$$

and hence $c_h\in[1.501360,\,1.501362]$.

The constant of Proposition 4.3 is $c_{2h}=\kappa_0 c_h^2$ with

$$\kappa_0=\prod_{p>2}\frac{1-2h(p)}{(1-h(p))^2}=\prod_{p\equiv 1\,(4)}\Big(1-\frac{(2p-1)^2}{(p^2-p+1)^2}\Big)\prod_{q\equiv 3\,(4)}\Big(1-\frac{1}{(q^2-q-1)^2}\Big),$$

since $h(p)/(1-h(p))=(2p-1)/(p^2-p+1)$ for split p and $h(q)/(1-h(q))=1/(q^2-q-1)$ for inert q. The factors are $1-O(p^{-2})$; the logarithms of the factors with $p>P$ are at most $4.01/p^2$ in absolute value, so the tail of $\log\kappa_0$ lies in $[-4.01/P,0]$ (the script uses the margin $8/P$). This gives $\kappa_0\in[0.738595,\,0.738597]$ and $c_{2h}\in[1.66485,\,1.66487]$. These computations are in floating point with explicit margins rather than in interval arithmetic; since c_h and c_{2h} enter only the value of the constant c_1 in Theorem 1.1, and not the positivity of $L-\lambda R$, this suffices. As rough independent checks, $(\log P)\prod_{2<p<P}(1-h(p))=1.50132$ and $(\log P)^2\prod_{2<p<P}(1-2h(p))=1.66477$, consistent with c_h, c_{2h} and with the $O(1/\log P)$ rate of convergence.

Data and code availability

The scripts behind Table 3 and Figures 2, 3 (`sup_error_scan.py`, `plot_sup_error.py`, `plot_error_hist.py`), Table 2 and Figure 4 (`certify_modular.py`, `plot_sieve_functions.py`), the choice of the sieve parameters and Figure 1 (`modular_sieve_scan.py`, `plot_sieve_landscape.py`) and the constants c_h, c_{2h} (`singular_constant.py`), together with their outputs, are available at

https://github.com/Ruqing1963/two-squares-theta-level

and archived at Zenodo, doi:10.5281/zenodo.23826176 (this version); version 2, with the bounds 9 and 6, is doi:10.5281/zenodo.23618710. The code and data of Part II, including the measurement of the norm and the exact check at the level 20, are at https://github.com/Ruqing1963/two-squares-mean-square. The numbers in the paper are those printed in the output files of the repository: Table 2 in `results/certify_modular_output_half.txt` (first line) and `results/certify_modular_output.txt` (lines 2–4); Table 3 and Figure 2 in `results/sup_error_scan_x21.txt` and `results/sup_error_scan_x23.txt`; the constants c_h, c_{2h} and c_1 in `results/singular_constant_output.txt`. The certification of Lemma 4.4 uses directed rounding; the experiments of Section 5 use floating point and are not certified.

Environment. The output files in the repository were produced with Python 3.12.4, `numpy` 1.26.4, `scipy` 1.13.1, `mpmath` 1.3.0 and `matplotlib` 3.8.4 under Windows 11; the versions are pinned in `requirements.txt`, and `README.md` lists the commands that regenerate every table and figure from scratch. Each script runs in a few minutes on a laptop.

What is rigorous and what is not. Three kinds of computation occur. (1) `certify_modular.py` (Table 2) uses interval arithmetic for the closed forms and directed rounding for everything else, as described in Appendix A; the six-decimal entries of Table 2 are the lines `L >=`, `R <=` and `certified: net = L - lam R >=` of its output file, and the two rational quantities are computed exactly. (2) `singular_constant.py` (the constants c_h, c_{2h}, c_1) uses floating point with explicit error margins (Appendix A.3); it affects only the value of the constant in Theorem 1.1, not the positivity of $L - \lambda R$. (3) `sup_error_scan.py` and the plotting scripts (Table 3, Figures 1–4) use floating point without error control; they provide evidence and illustration, and nothing in the proofs depends on them.

During the preparation of this work the author used the Claude language model and the `mpmath` Python library for the computer-assisted interval arithmetic verification of Section 4 and for the numerical experiments of Section 5. All analytical derivations and proofs were independently audited and verified by the author.

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